

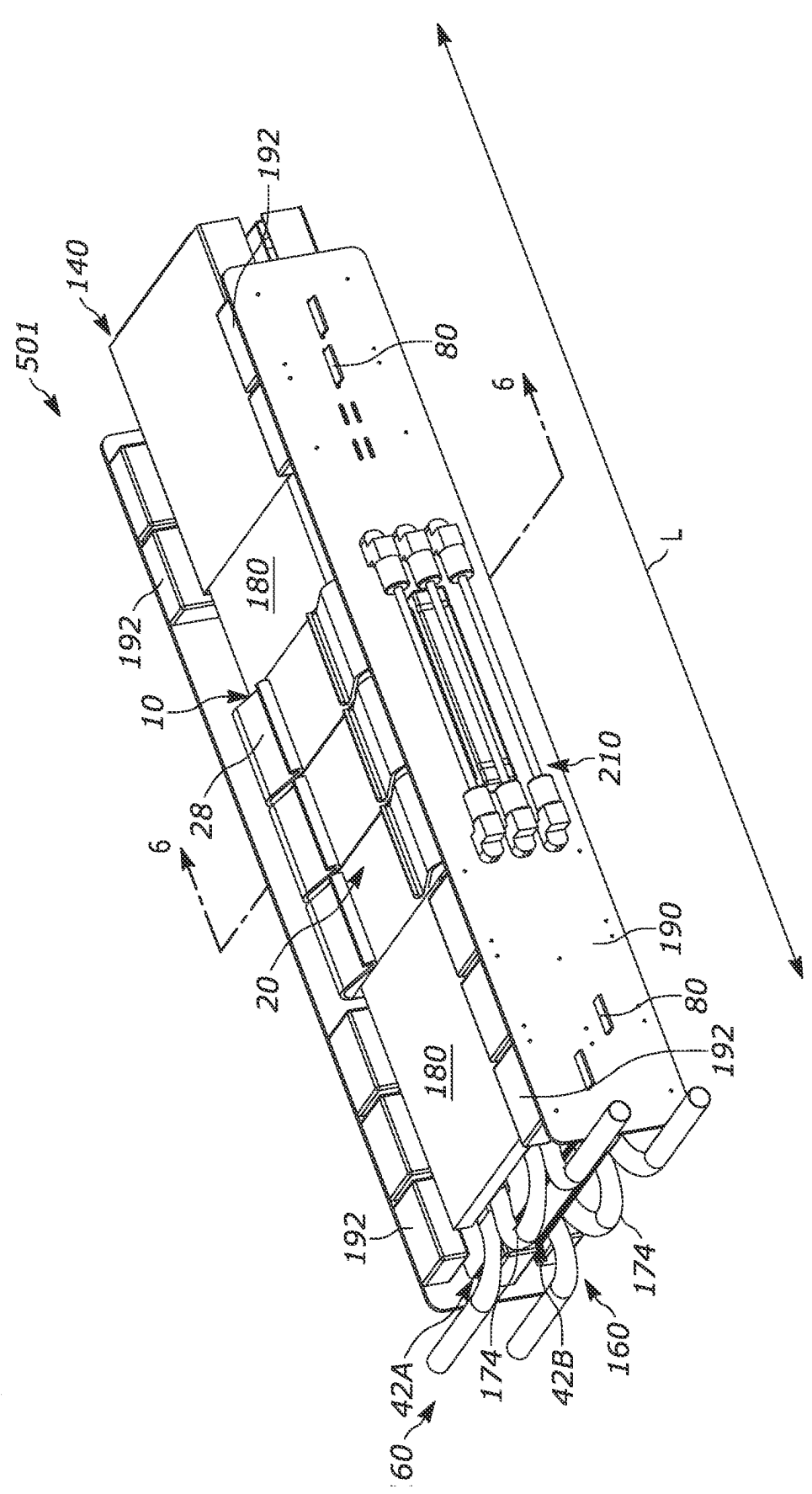


FIG. 1

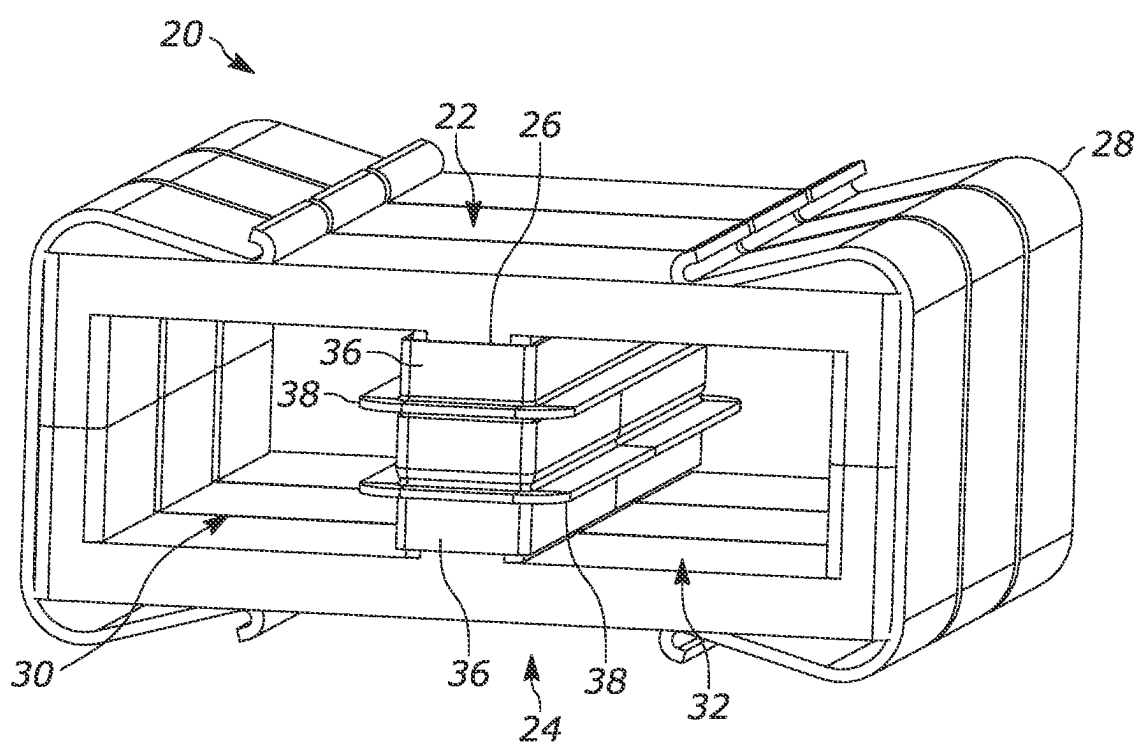


FIG. 2

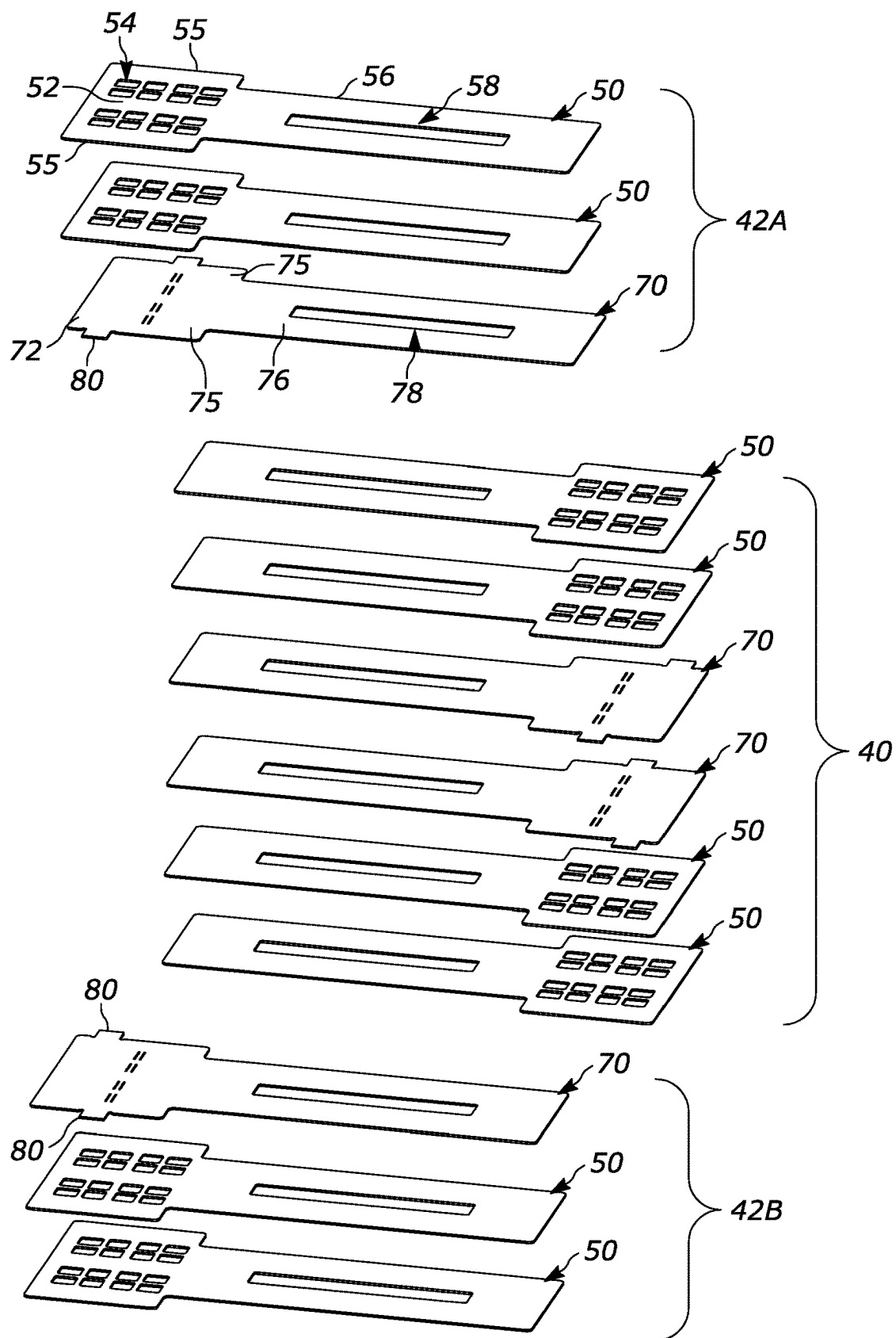


FIG. 3

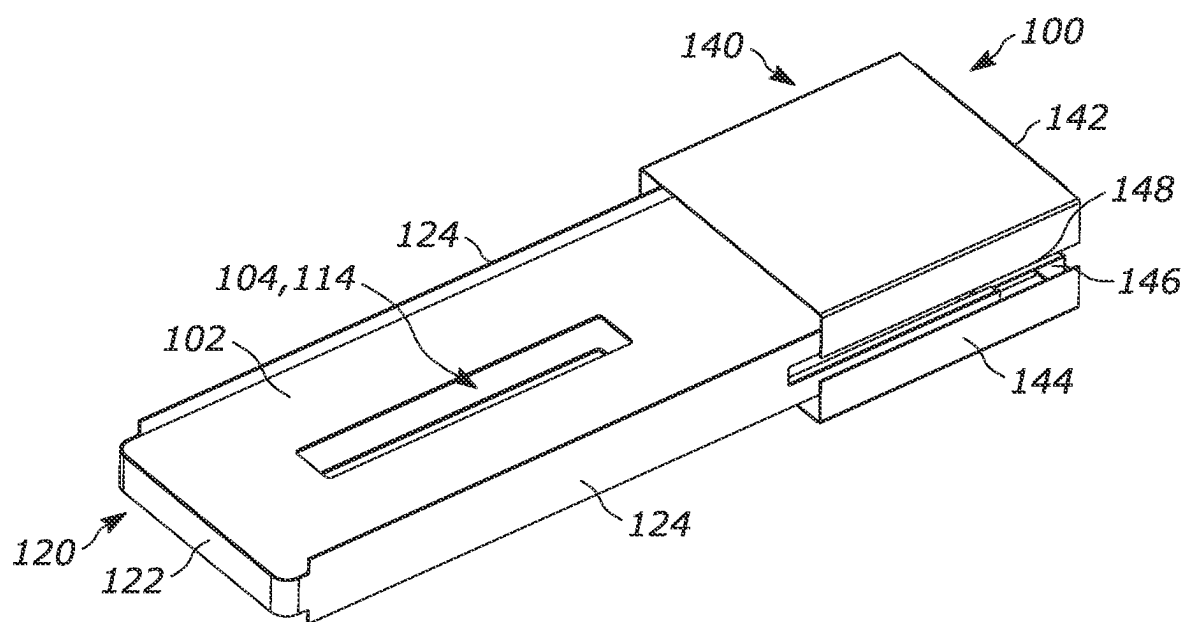


FIG. 4A

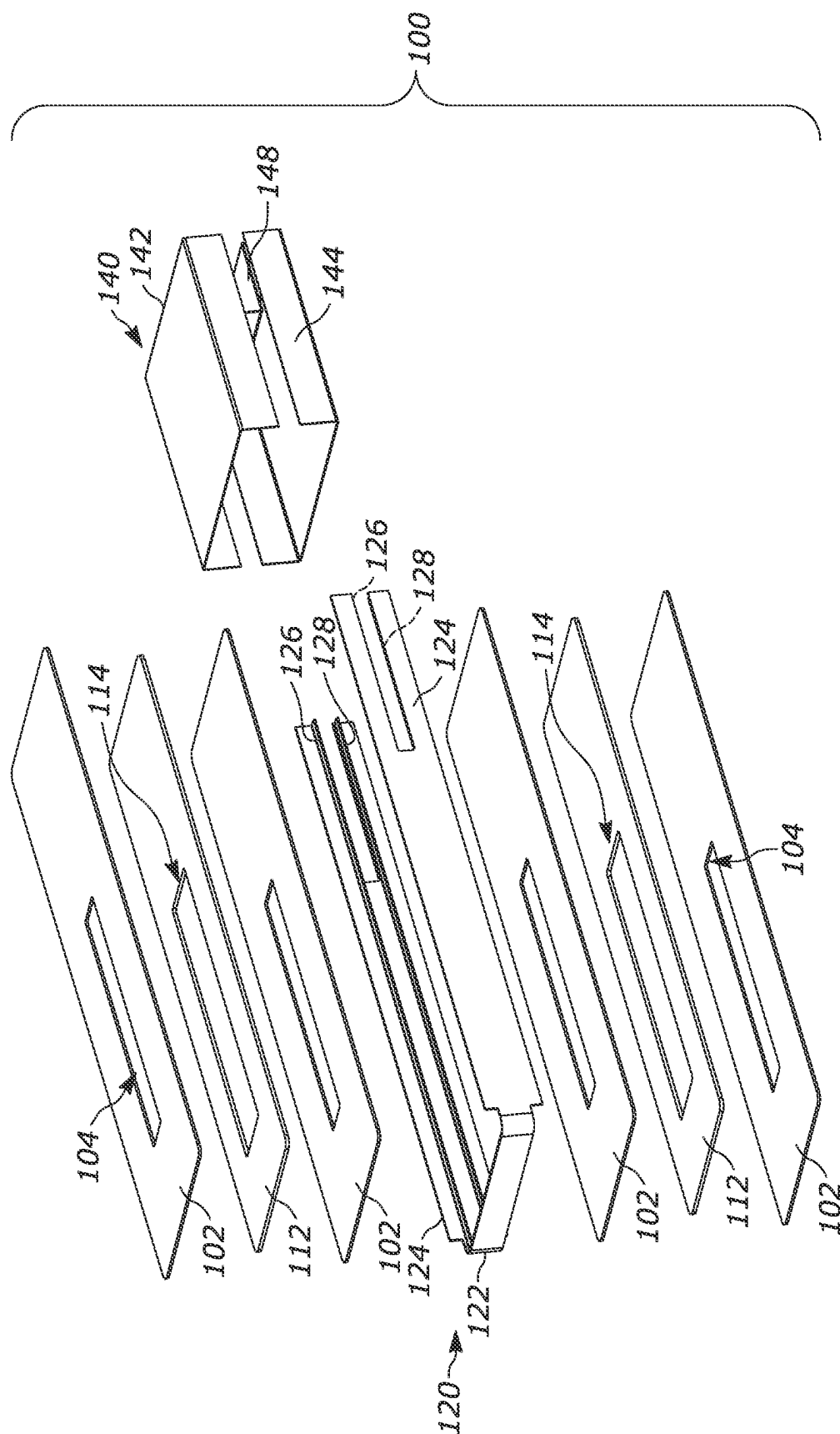


FIG. 4B

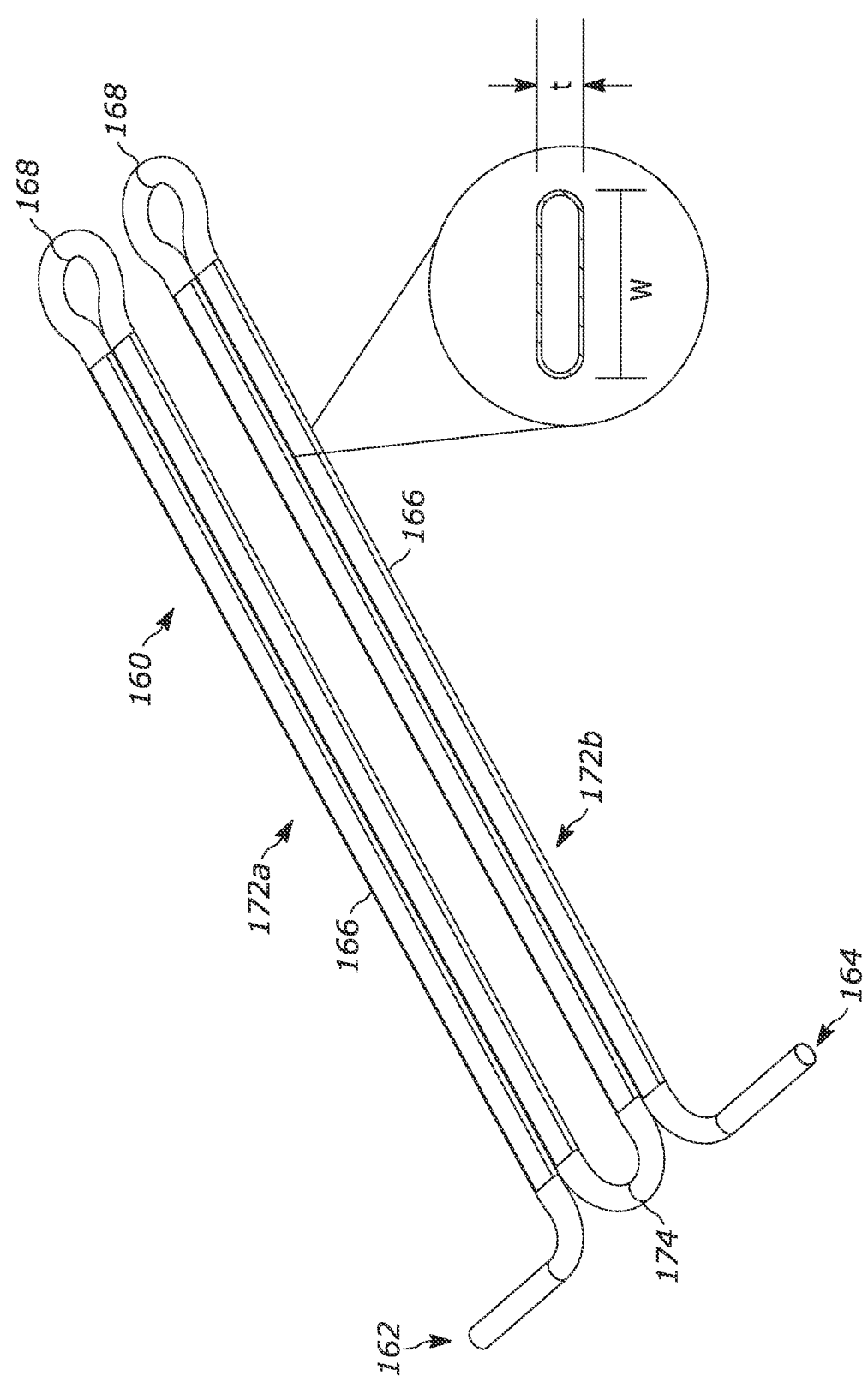


FIG. 5

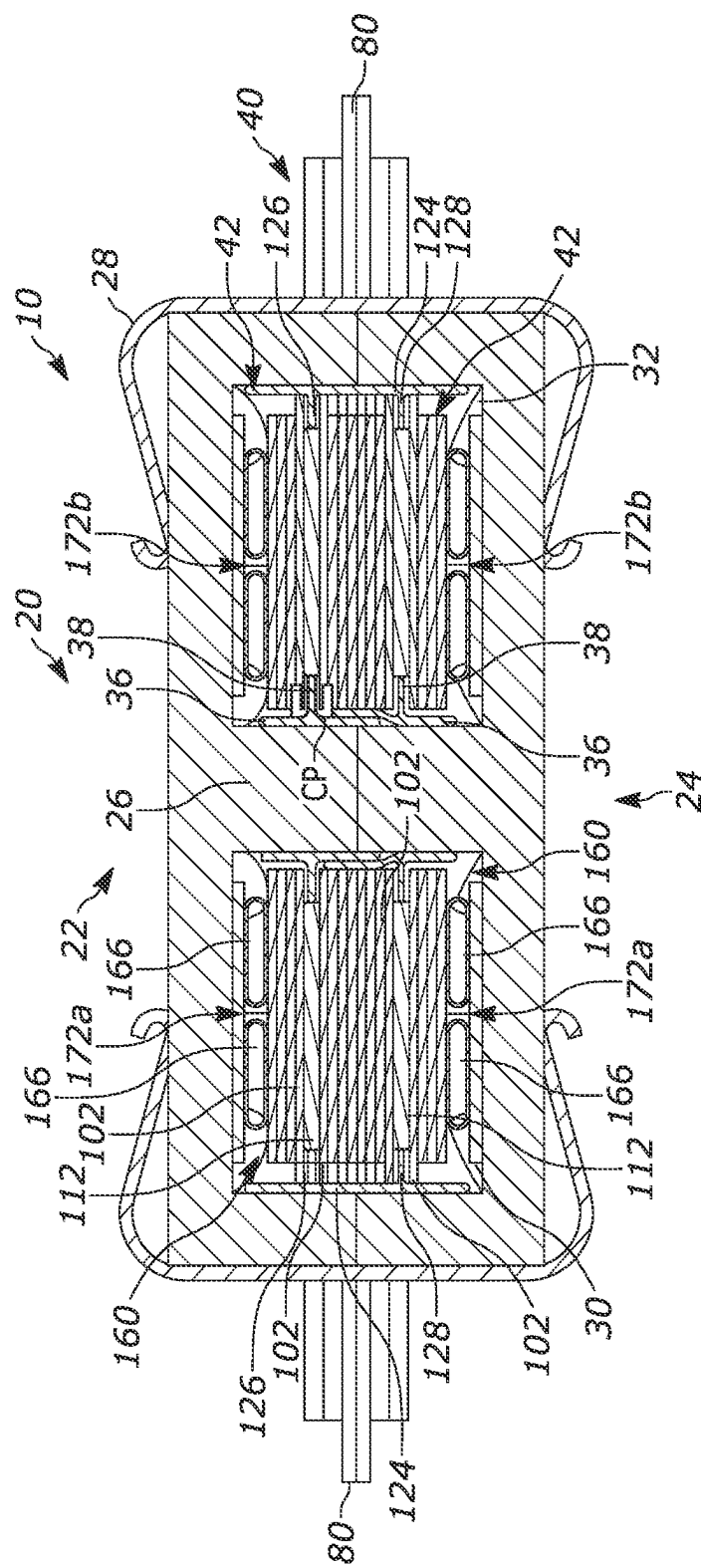


FIG. 6

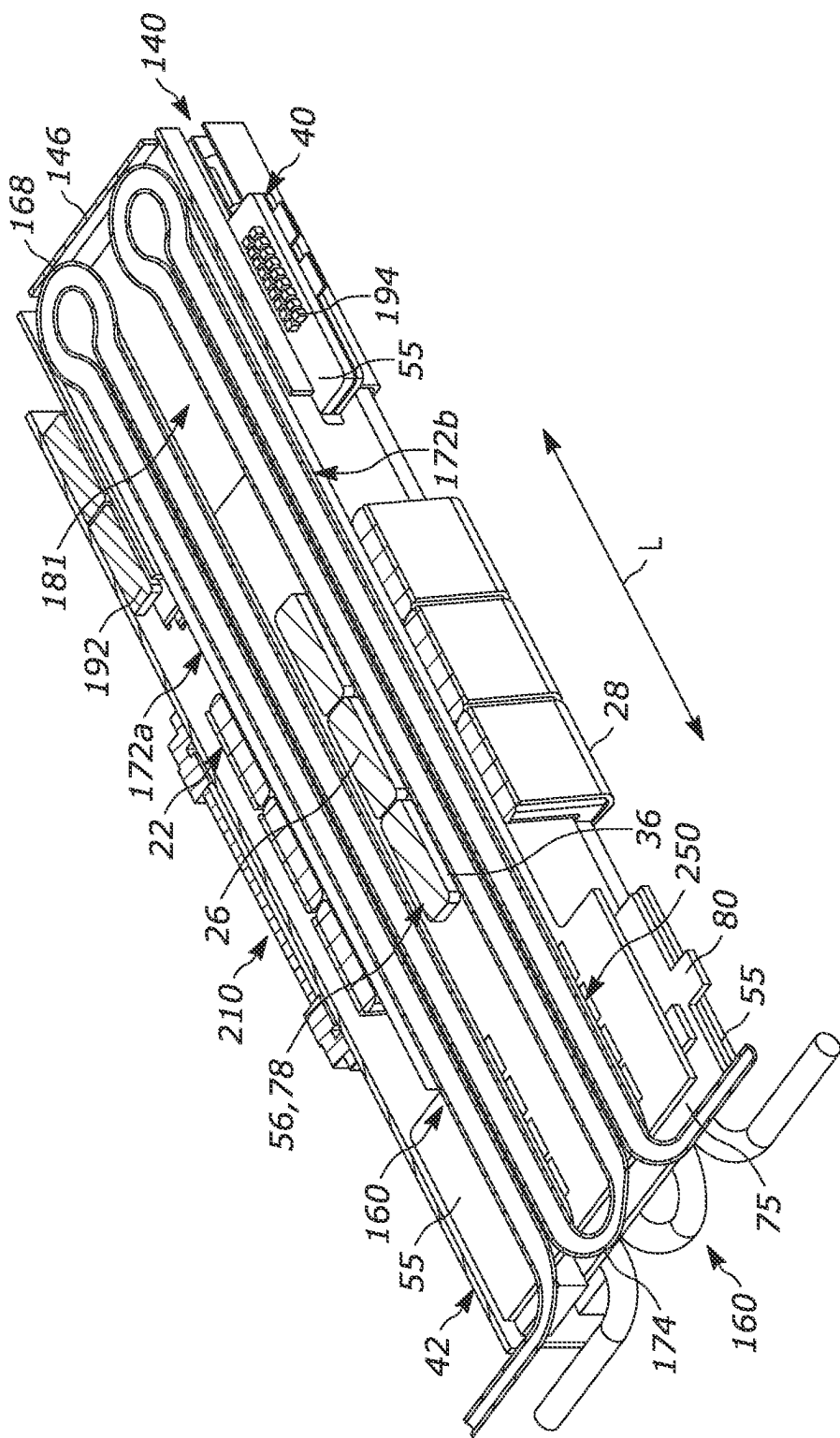
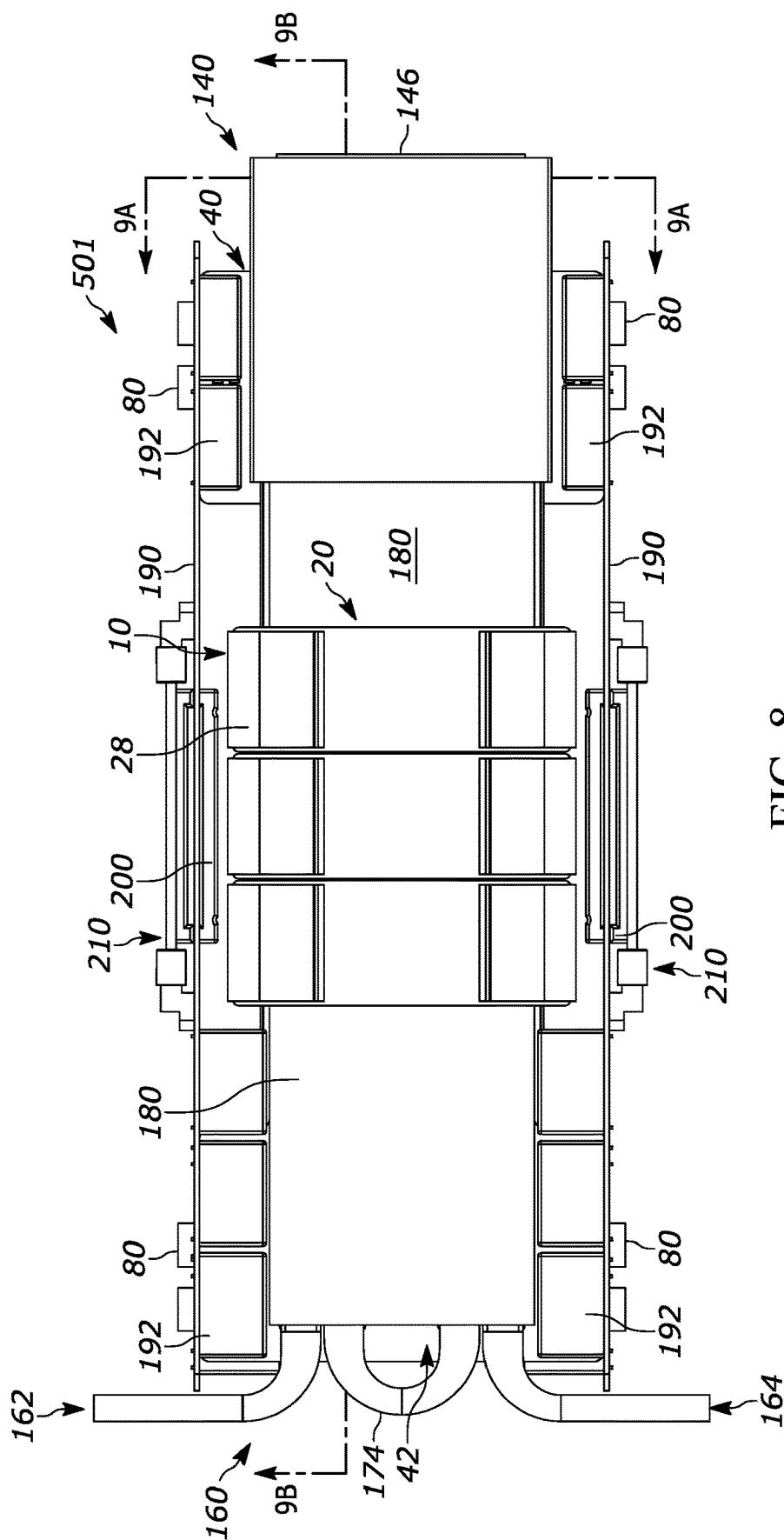


FIG. 7



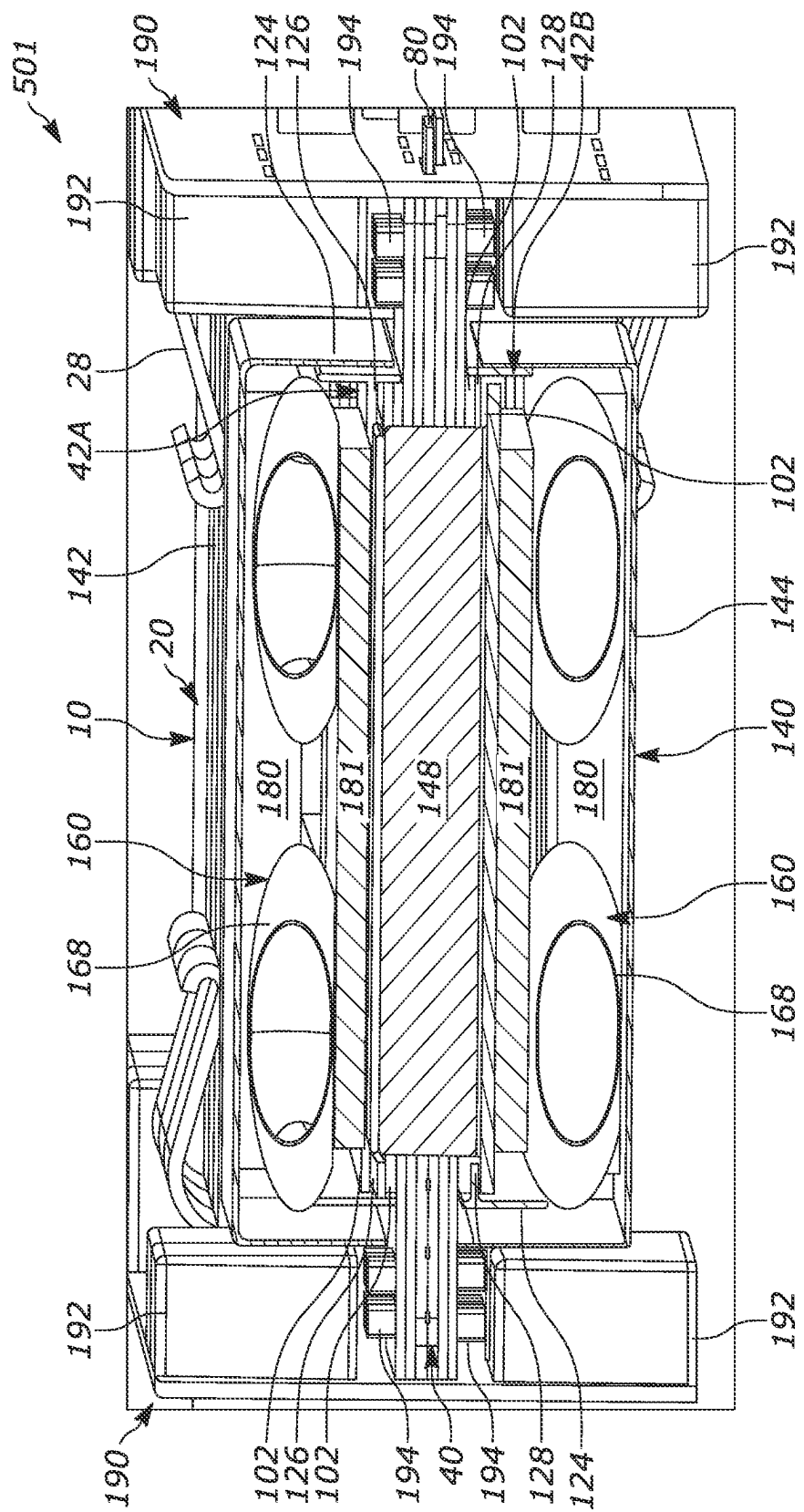


FIG. 9A

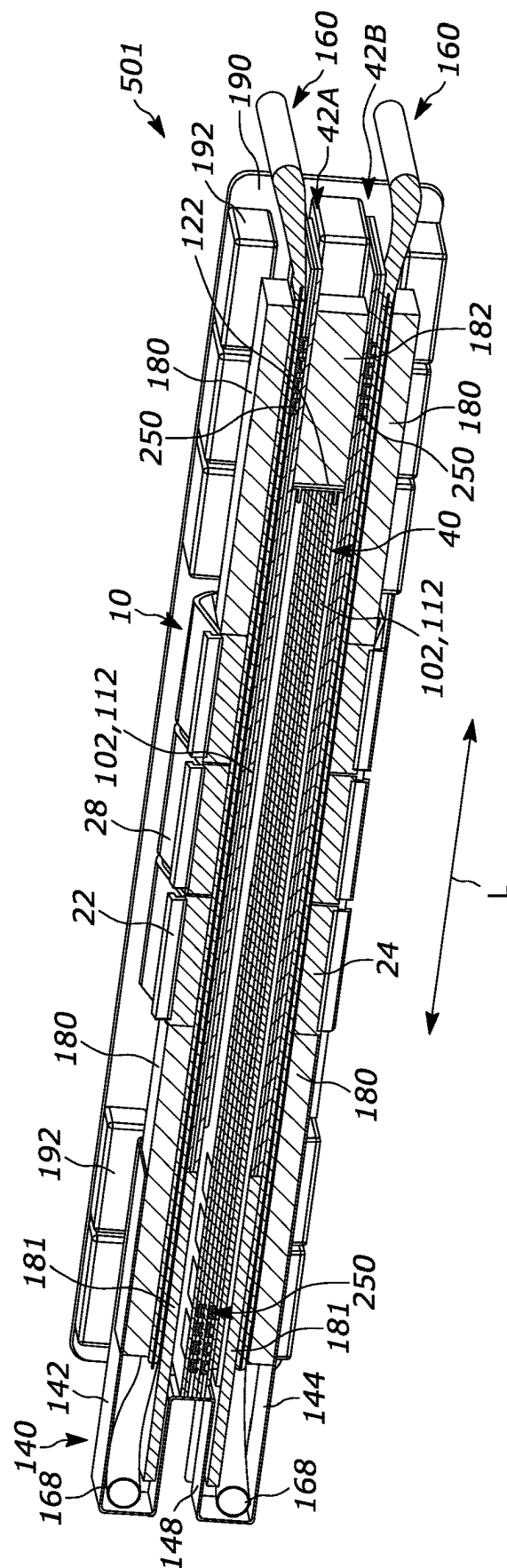


FIG. 9B

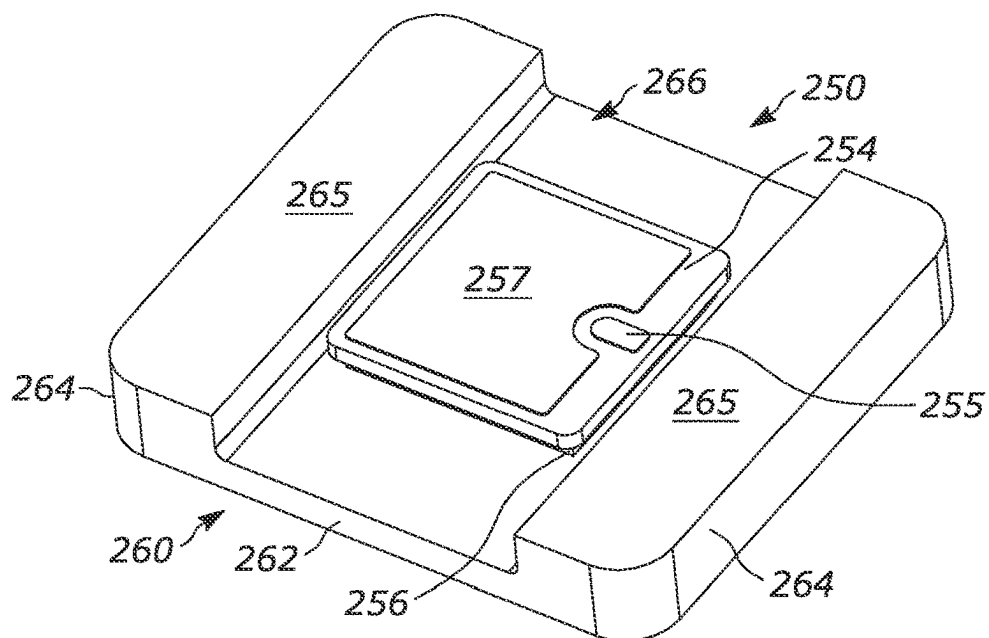


FIG. 10A

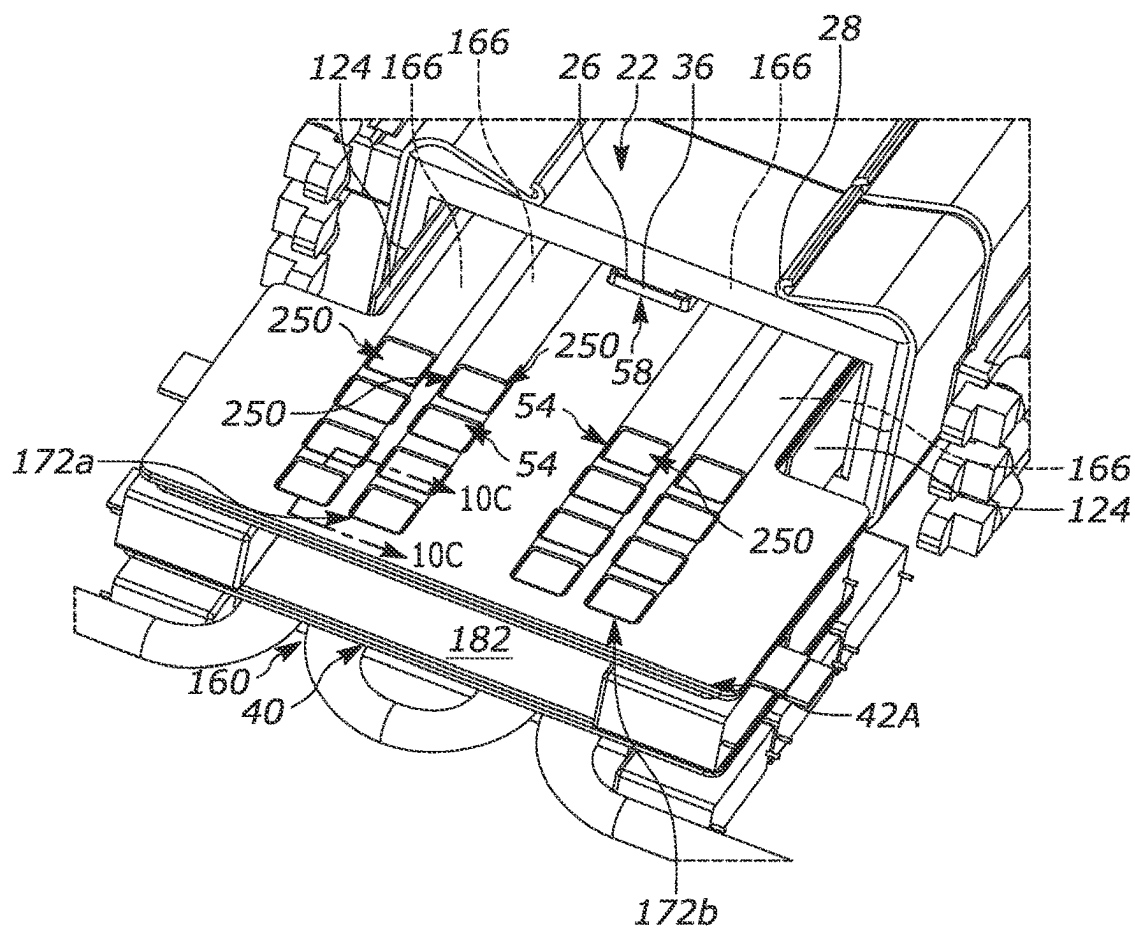


FIG. 10B

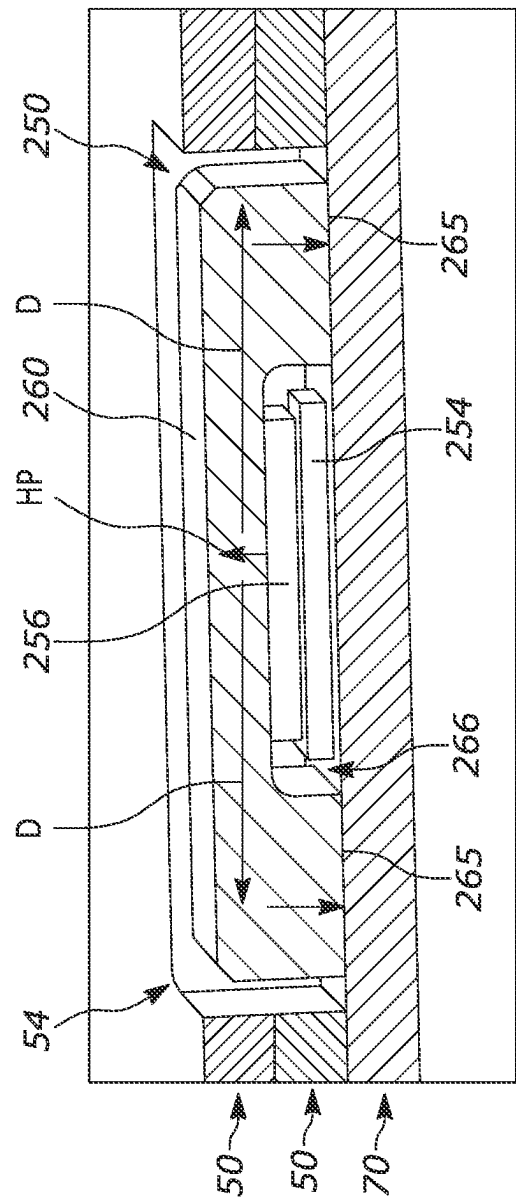


FIG. 10C

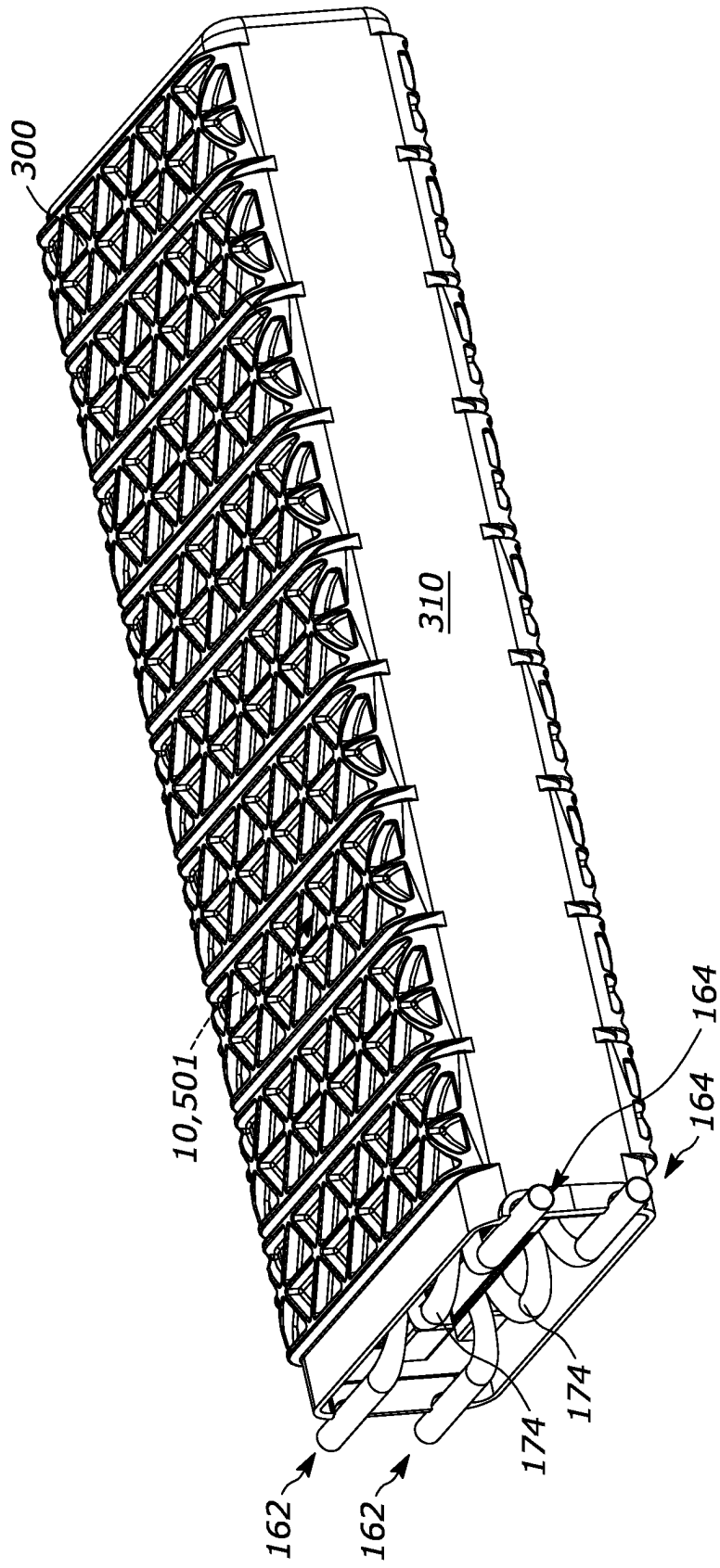


FIG. 11

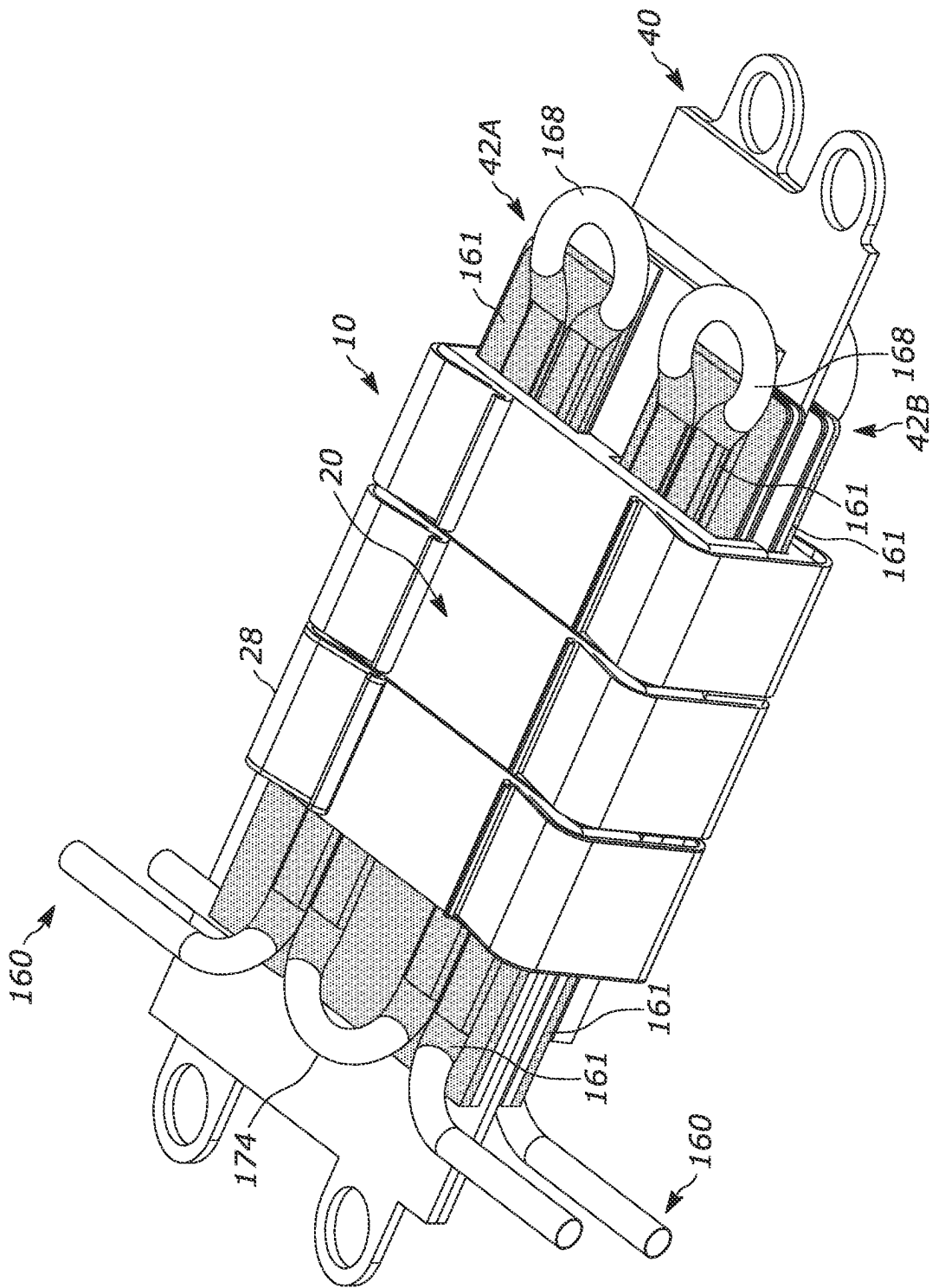


FIG. 12A

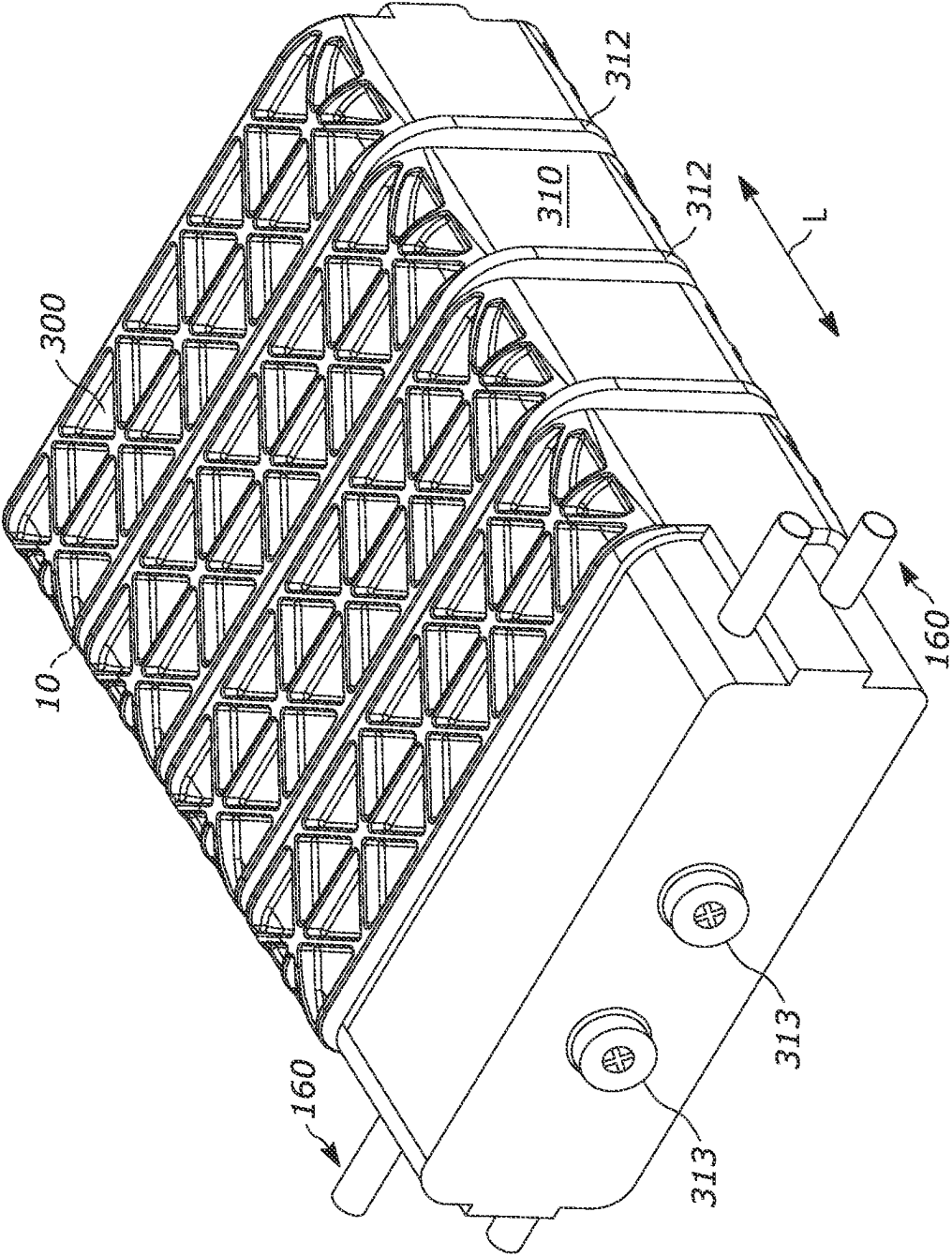


FIG. 12B

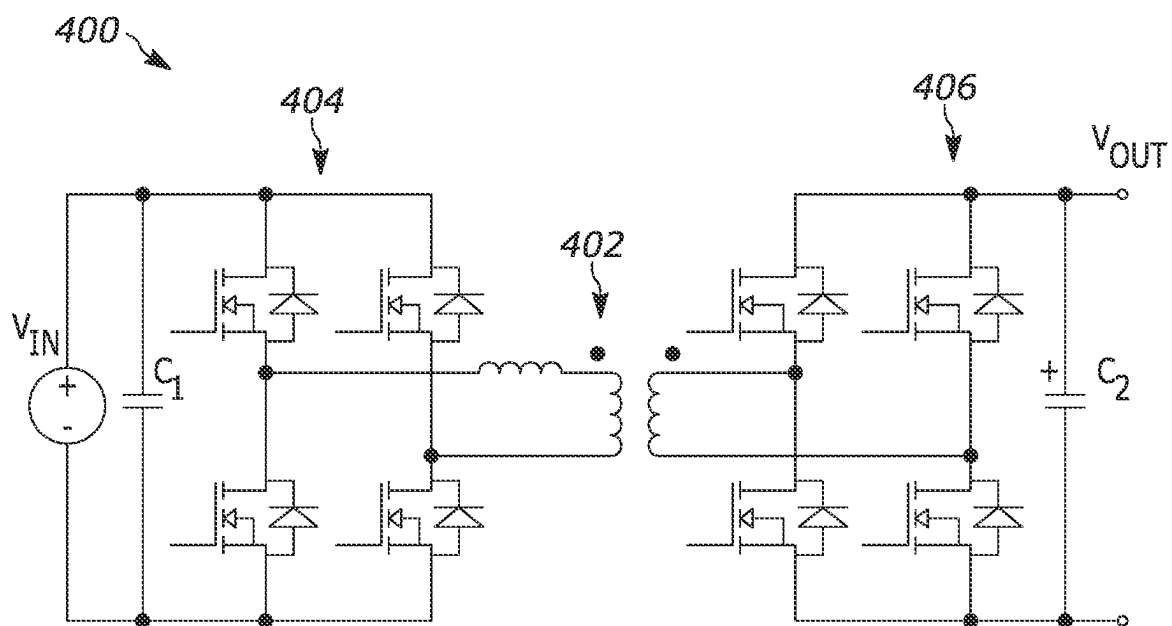


FIG. 13

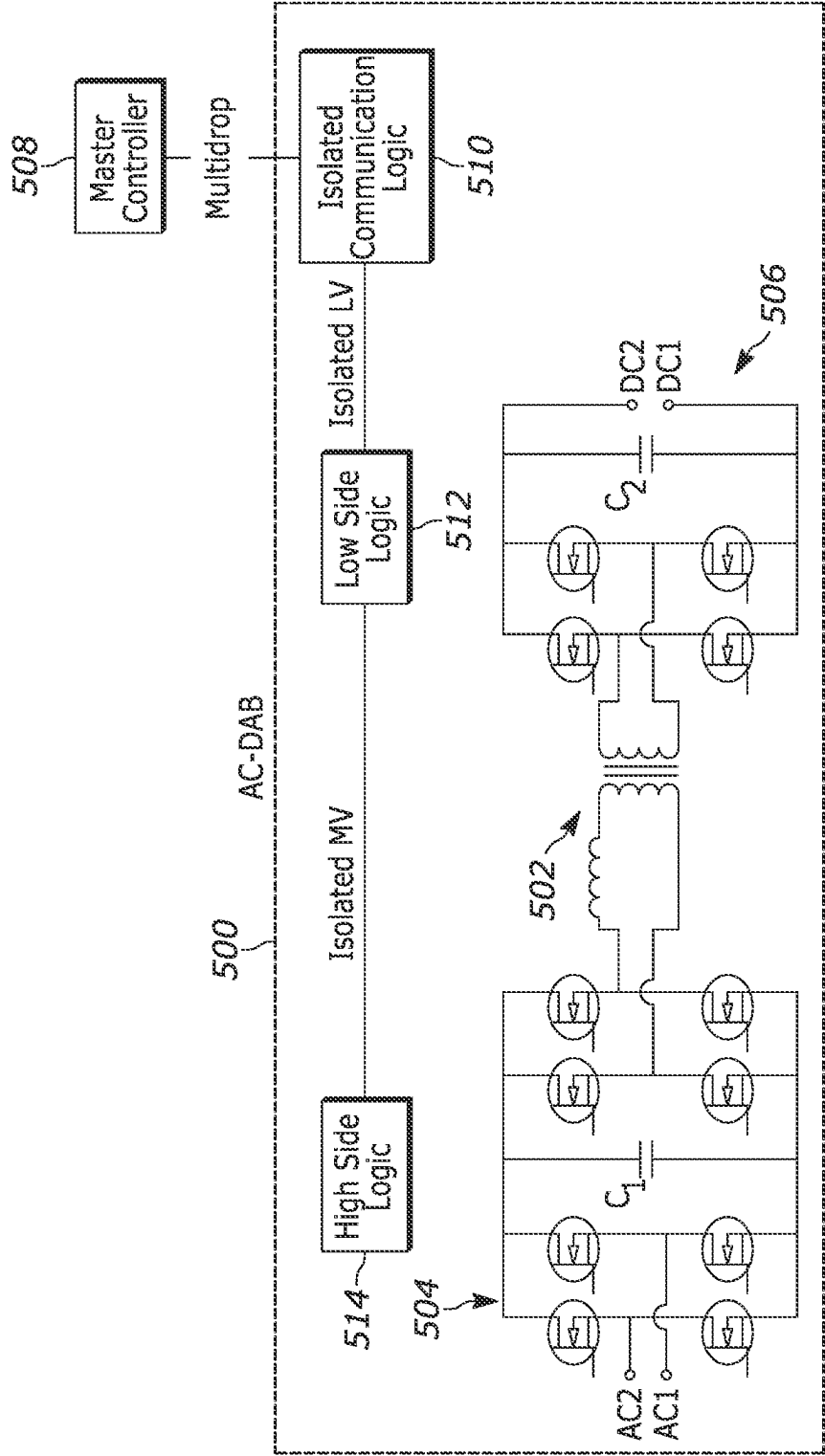


FIG. 14

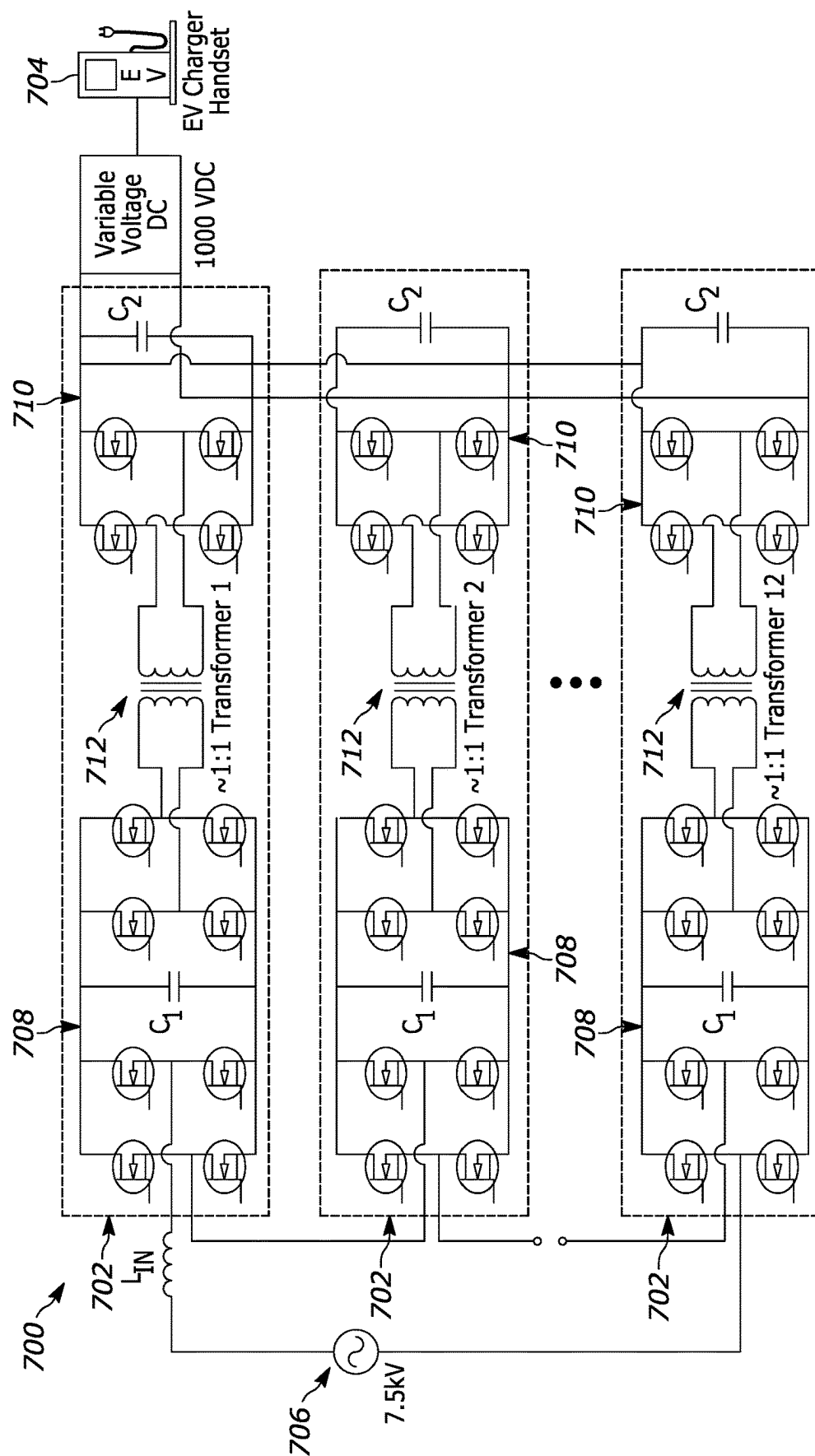


FIG. 15

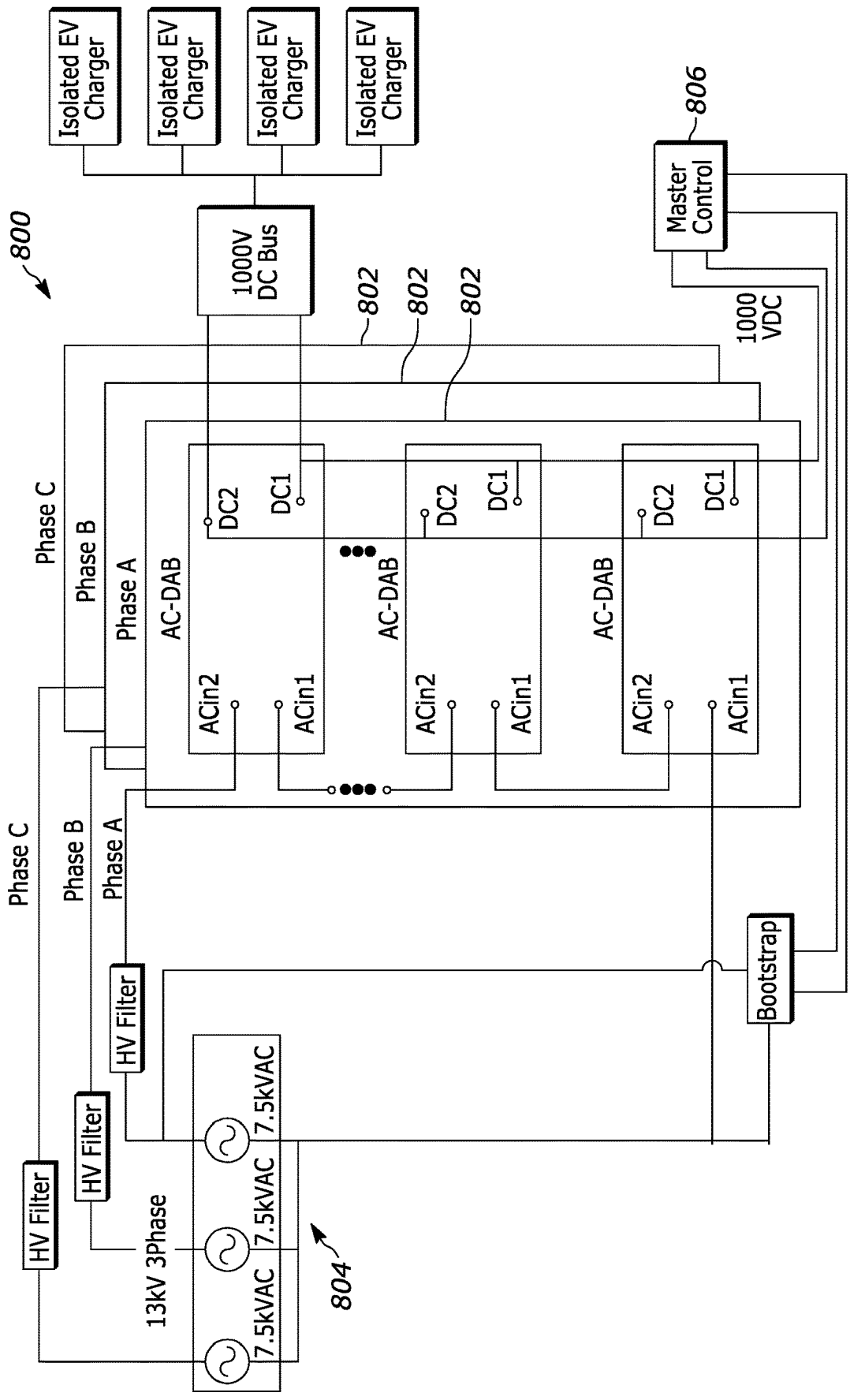


FIG. 16

PLANAR TRANSFORMER AND DUAL ACTIVE BRIDGE

TECHNICAL FIELD

[0001] The present invention relates generally to transformers, and specifically to a planar, high power, medium voltage transformer and an associated dual-active bridge module.

BACKGROUND

[0002] Medium voltage AC to DC converters can be used in electric vehicle (EV) charging stations. More specifically, medium-voltage AC to ~1000 VDC conversion is accomplished using many identical, high-frequency transformer-based AC to DC conversion modules connected in series on the AC medium voltage side and connected in parallel on the DC side—a so-called series-in parallel-out architecture. By placing the primary windings in series and the secondary windings in parallel, a large voltage step down can be achieved with transformers that are 1:1 or nearly 1:1 turns ratio. The transformers can achieve a frequency of about 20 kHz-200 kHz.

[0003] Due to this construction, however, the eddy current and AC losses can be high. As a result, the conductors in the transformer tend to be made from LITZ wire, laminated tape or arranged in a planar transformer. LITZ wire has fine strands of conductors that are insulated from each other but can be expensive. Moreover, the insulation used with LITZ wire tends to occupy a large percentage of the winding space. Laminated tape is thin and provides wide layers of conductors separated by insulation, which is difficult to implement in standard cores because edge winding is problematic with laminated tape.

[0004] A second challenge is the high voltage isolation that is needed between the primary and secondary windings in the transformer. Because the primary windings are many different voltages and the secondary voltages are similar, significant isolation is required. This can require 30-50 kV isolation for 15 kV class systems. Further, these devices require a Basic Impulse Level (BIL) of over 100 kV. These isolation voltages require either very large creepage and clearance distances or continuous solid insulation between the layers. Additionally, there is a desire to have the conductive components for transformer cores and for cooling to be ground referenced and isolated from any of the winding voltages.

[0005] A third challenge is the minimization of parasitic resistances, capacitances, and inductances in locations that make it difficult to switch at high frequencies and that contribute to heat generation that needs to be removed and reduce efficiencies. While some inductance is necessary for Zero Voltage Switching (ZVS) operation, too much results in excess reactive current in the bus capacitors, as well as additional losses in snubbers. Capacitance within the winding makes it more challenging to achieve ZVS switching and capacitance across the ground wall increases losses in the switches, and contributes to EMI issues.

[0006] A fourth challenge is to keep the transformer cool while achieving the above three requirements. If liquid cooling or heat pipes are employed, placing metal in or near the transformer can have high eddy current losses depending on how it is arranged, how conductive the metal is, and how thick and what orientation the metal placed. Further, the

transistors added on each side of the transformer have similar requirements of low parasitics, good cooling, and high isolation.

SUMMARY

[0007] A high-frequency planar transformer construction for high-power medium-voltage applications is described. The transformer includes a magnetic core, planar windings composed of one or more printed circuit boards (PCBs), an embedded cooling tube, and insulation layers that are substantially ceramic. Further embodiments include power transistors that switch current and control power flow through the device on each of the primary and secondary sides of the transformer to implement a compact power-transfer circuit. The transistors on each side of the transformer are attached to the same PCB and share the same insulation structures as at least one of the respective transformer windings. The embedded cooling tube could also be a heat pipe. Further embodiments package the transistors as bare die in a transistor die assembly that reduces the thermal resistance between the bare die and the coolant.

[0008] In one example, a planar transformer includes a primary winding having one or more PCBs each including metallic traces. A secondary winding includes one or more PCBs having metallic traces. A ceramic electrical insulator electrically separates the primary and secondary windings. A magnetic core extends entirely around the windings and the electrical insulator.

[0009] In another example, a dual active bridge (DAB) module includes a planar transformer having a first winding and a second winding and a magnetic core. The first winding is formed as metallic traces on at least one first printed circuit board (PCB). The second winding is formed as metallic traces on at least one second PCB. The magnetic core being formed around at least a portion of the first and second PCBs. A first plurality of transistors is formed on at least one of the first PCBs to form a first stage. The first transistors are controlled to provide a first current through the first winding based on an input voltage to provide a second current in the second winding. A second plurality of transistors is formed on at least one of the second PCBs to form a second stage. The second transistors are controlled to provide an output voltage based on the second current.

[0010] In another example, a method for providing power to electric vehicles (EVs) at an EV charging station includes electrically coupling input stages of a respective plurality of dual active bridge (DAB) modules, each of the DAB modules being formed from a plurality of printed circuit boards (PCBs) and having a respective one of the input stages, a planar transformer comprising a primary winding and a secondary winding, and an output stage. The output stages of the respective DAB modules are electrically coupled, such that the DAB modules are arranged as a voltage converter circuit. An AC current is provided to the voltage converter circuit via the input stages of the DAB modules. Switches associated with the input stages of the respective DAB modules are controlled to provide a primary current through the primary winding of each of the planar transformers of each of the DAB modules to provide a secondary current in the second winding of each of the planar transformers of each of the DAB modules. Switches associated with the output stages of the respective DAB modules are controlled to provide an output voltage from each of the DAB modules based on the respective secondary current

provided in the secondary winding of the planar transformer of each of the DAB modules. The output voltage is provided to each of a plurality of isolated EV chargers for the EV charging station.

[0011] In another example, a voltage converter circuit includes a plurality of dual active bridges (DABs). Each DAB has a first stage with a plurality of first switches, a second stage with a plurality of second switches, and a transformer having a first winding coupled to a least one of the first switches and a second winding coupled to at least one of the second switches. At least one of the first and second stages of a plurality of the DABs that is a proper subset of the DABs are arranged in parallel with respect to the first winding of the transformer in each of the respective plurality of the DABs, with the parallel first stages of the plurality of the DABs being arranged in series with respect to the first winding of the transformer of at least one other DAB of the DABs, or at least one of the first and second stages of a plurality of the DABs that is a proper subset of the DABs are arranged in series with respect to the second winding of the transformer in each of the respective plurality of the DABs, with the series second stages of the plurality of the DABs being arranged in parallel with respect to the second winding of the transformer of at least one other DAB of the DABs.

[0012] Other objects and advantages and a fuller understanding of the invention will be had from the following detailed description and the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 is a schematic illustration of an example dual active bridge (DAB) module having a planar transformer in accordance with an aspect of the present invention.

[0014] FIG. 2 is a perspective view of a magnetic core of the transformer.

[0015] FIG. 3 is an exploded view of printed circuit boards (PCBs) for forming primary and secondary windings of the transformer.

[0016] FIG. 4A is a perspective view of an insulating assembly for the transformer.

[0017] FIG. 4B is an exploded view of the insulating assembly of FIG. 4A.

[0018] FIG. 5 is a perspective view of a cooling structure for the transformer.

[0019] FIG. 6 is a section view taken along line 6-6 of FIG. 1.

[0020] FIG. 7 is a top view of the DAB module with components removed.

[0021] FIG. 8 is a top view of the DAB module.

[0022] FIG. 9A is a section view taken along line 9A-9A of FIG. 8.

[0023] FIG. 9B is a section view taken along line 9B-9B of FIG. 8.

[0024] FIG. 10A is a perspective view of a transistor assembly package of the DAB module.

[0025] FIG. 10B is a top view of a portion of the DAB module with an array of transistors.

[0026] FIG. 10C is a section view taken along line 10C-10C of FIG. 10B.

[0027] FIG. 11 is a schematic illustration of the DAB module enclosed in potting.

[0028] FIG. 12A is a schematic illustration of a stand-alone planar transformer.

[0029] FIG. 12B is a schematic illustration of the stand-alone transformer enclosed in potting and a case.

[0030] FIG. 13 is an example of a circuit diagram of a dual active bridge (DAB) circuit.

[0031] FIG. 14 is another example of a circuit diagram of a DAB circuit.

[0032] FIG. 15 is an example of a voltage converter circuit.

[0033] FIG. 16 is another example of an electric vehicle (EV) charging system.

DETAILED DESCRIPTION

[0034] The present invention relates generally to transformers, and specifically to a planar, high power, medium voltage transformer and an associated DAB module, i.e., included in the same package assembly as a DAB module. FIGS. 1-10C illustrate an example DAB module 501 having a planar transformer 10 and extending in a direction of length L in accordance with the present invention. It is appreciated that the planar transformer 10 can alternatively be provided in other, non-DAB modules or circuits or be stand-alone, as will be described.

[0035] As shown in FIG. 2, the transformer 10 includes a core 20 formed from a pair of cooperating halves 22, 24. The halves 22, 24 can be configured as E-cores or E-I-cores. Regardless, the core 20 is formed from a magnetic material, such as ferrite. In one example, the core 20 is a high frequency ferrite material, such as manganese zinc ferrite, nickel zinc ferrite or other ferrite suited for an operation frequency of about 20 kHz-200 kHz.

[0036] As shown, the halves 22, 24 are E-cores that each includes a central support member 26. The support members 26 from each half 22, 24 engage one another and cooperate to define a pair of windows or passages 30, 32 in the core 20.

[0037] A pair of bobbins 36 encircle the respective support members 26 and are stacked atop one another so as to extend the entire collective height of the support members. Each bobbin 36 is generally tubular and includes a radially extending shelf or flange 38 that encircles the associated support member 26. The bobbins 36 are made from an electrically insulating material, such as nylon, silicone, PPS, PPA, PBT or polyester. The bobbins 36 should also have a Comparative Tracking Index (CTI) of greater than 600V and high temperature capability.

[0038] The core 20 shown includes three associated pairs of halves 22, 24 aligned along the length L. It will be appreciated, however, that more or fewer associated pairs of halves 22, 24 can be provided. Regardless, a compression clip 28 is provided for each associated pair of halves 22, 24 for biasing the halves towards one another. That said, three compression clips 28 are provided on the example core 20 shown. It will be appreciated that alternative/additional means could be used to secure the core 20, e.g., adhesive, tape, other fasteners, etc.

[0039] Referring to FIG. 3, the transformer 10 further includes a primary winding 40 and a secondary winding 42 formed from at least one secondary winding portion. In one example shown, a pair of secondary winding portions 42A, 42B is provided on opposite sides of the primary winding 40 and cooperate to define the secondary winding 42. In any case, each winding 40, 42 can be formed from one or more printed circuit boards (PCB) having either a first type 50 or a second type 70. In the example shown, each winding 40, 42 is formed from stacked PCBs 50, 70. It will be appreci-

ated, however, that each PCB 50, 70 could be formed as a single layer or multiple layers and that the PCBs forming each winding 40, 42 could be single layers PCBs, multiple layer PCBs, combinations thereof, and/or combinations of PCBs having different number of layer(s) from one another within or between the windings 40, 42.

[0040] Regardless, the first PCB 50 includes a base 52 having an array of openings 54 extending through the thickness thereof. Tabs 55 extend outward from opposite sides of the base 52. A projection 56 extends longitudinally away from the base 52. An elongated opening 58 extends along the length of the projection 56 and passes entirely therethrough.

[0041] The second PCB 70 includes a base 72 and tabs 75 extending outward from opposite sides of the base. A projection 76 extends longitudinally away from the base 72. An elongated opening 78 extends along the length of the projection 76 and passes entirely therethrough. Support members 80 extend outward from opposite sides of the tabs 75.

[0042] Each PCB type 50, 70 is formed from a dielectric material. Electrical conductors or traces (not shown) are etched into a side (the top side as shown) of each PCB 50, 70 and generally encircle the respective opening 58, 78. The etched traces can extend one or more times around the respective openings 58, 78. The conductors can be made of copper or any other electrically conductive material and insulated with an epoxy-glass composite, such as FR-4. Fillers can be used in the insulation to enhance thermal conductivity.

[0043] As an example, the electrical conductors of at least one of the primary winding 40 and the secondary winding 42 can include multiple conductors arranged in parallel, such as to reduce eddy current losses. Additionally, for example, the conductor paths of the parallel conductors provided for the primary winding 40 and/or the secondary winding 42 can be transposed to link substantially equal magnetic flux of the planar transformer 10 to improve current sharing between the parallel conductors.

[0044] As shown, PCBs 50, 70 are stacked on one another to form the primary winding 40, with a pair of the PCBs 70 being sandwiched between pairs of the PCBs 50 in a six-layer construction. It will be appreciated that more or fewer of either PCBs 50, 70 can be provided to form the primary winding 40. The PCBs 50, 70 are bonded to one another with solder or sintering to form the 42A, 42B primary winding 40. Regardless, the openings 58, 78 are aligned with one another and the bases 52, 72 are aligned with one another. The PCBs 50, 70 can be bonded to one another with solder or sintering to form the primary winding 40.

[0045] Furthermore, as shown, PCBs 50, 70 are stacked on one another to form each secondary winding portion 42A, 42B, with a pair of the PCBs 50 being positioned on one side of a PCB 70 in a three-layer construction. It will be appreciated that more or fewer of either PCB 50, 70 can be provided to form the secondary winding portions 42A, 42B, but the secondary winding portions have the same configuration as one another. Regardless, the openings 58, 78 are aligned with one another and the bases 52, 72 are aligned with one another. The windings 40, 42 are oriented such that the aligned bases 52, 72 of the primary winding 40 is at an opposite end of the stack of PCBs than the aligned bases 52, 72 on both secondary winding portions 42A, 42B.

[0046] Insulation formed as an electrical insulator or insulating assembly 100 is also provided (FIGS. 4A-4B) for helping to electrically insulate the primary winding 40 from each secondary winding portion 42A, 42B. That said, components of the insulating assembly 100 are formed from an electrically insulating material, such as a ceramic. Example ceramic materials include, aluminum (III) oxide (Al_2O_3), aluminum nitride, silicon nitride, and boron nitride. As an example, the ceramic material(s) selected for the insulating assembly 100 can have dimensions that are suitable for providing a nominal operating potential between two of the electrical components, e.g., between the primary side and the secondary side of the transformer 10, of at least 2 kV.

[0047] The insulating assembly 100 includes elongated first insulating sheets 102 each having an elongated opening 104 extending along the length thereof. Second insulating sheets 112 each have an elongated opening 114 extending along the length thereof. The second insulating sheets 112 are smaller in both length and width than the first insulating sheets 102, i.e., the insulating sheets are different sizes. An edge insulator 120 has a generally U-shaped configuration and includes a base 122 and a pair of legs 124 extending therefrom parallel to one another. A pair of support rails 126, 128 extends along the entire length of the edge insulator 120 and radially inward. The support rails 126, 128 extend parallel to one another. An end cap insulator 140 has a generally box-shaped construction and includes a pair of C-shaped cover members 142, 144 connected together by a common end wall 146. A C-shaped projection 148 extends from the wall 146 into the interior between the cover members 142, 144.

[0048] When the insulating assembly 100 is assembled, two first insulating sheets 102 are positioned on opposite sides of the support rail 126 and abut both the support rail and the base 122. A single second insulating sheet 112 is positioned between the first insulating sheets 102 and within the same plane as the support rail 126. Similarly, two first insulating sheets 102 are positioned on opposite sides of the support rail 128 and abut both the support rail and the base 122.

[0049] A single second insulating sheet 112 is positioned between the first insulating sheets 102 and within the same plane as the support rail 128. This aligns all the openings 104, 114 of the sheets 102, 112 with one another. Due to this configuration, a three-piece or three-layer insulating member is associated with each rail 126, 128 and constructed from a pair of first insulating layers 102 on opposite sides of a single second insulating layer 112. The end cap insulator 140 extends over the free ends of the legs 124.

[0050] Turning to FIG. 5, a tubular cooling structure 160 extends from a first end 162 to a second end 164. Straight portions 166 extend parallel to one another and are connected at their terminal ends by curved portions 168. In one example, the straight portions 166 have flattened oval or generally rectangular cross-sections so as to increase the width (w) and decrease in thickness (t) compared to a rounder cross-section. The thickness (t) is on the order of about 0.05-1.0 mm, in particular about 0.1-0.30 mm. As will be discussed, the cooling structure 160 is configured to receive a cooling medium.

[0051] In any case, the portions 166, 168 are configured to define multiple out-and-back passes of the cooling structure 160, such as the pair of passes 172a, 172b shown. The passes 172a, 172b are connected end-to-end by a connecting por-

tion 174. The number of passes 172 is even to position the ends 162, 164 on the same side of the cooling structure 160. The cooling structure 160 is made of metal, such as stainless steel including a 300 series or austenitic stainless steel.

[0052] It will be appreciated that the cooling structure 160 could instead be formed as a metallic, e.g., stainless steel, heat pipe. In this configuration, the heat pipe has capillary structure and a working fluid, e.g., methanol, therein that boils at approximately the working temperature of the windings 40, 42. The heat pipe can have a rounded profile or the flattened profile exhibited by the cooling structure 160 shown. The heat pipe can have the same contour/shape as the cooling structure 160 shown, i.e., multiple passes 172a, 172b, or be configured with more or fewer passes. In one example, the heat pipe includes a single straight section 166, and one or more heat pipes can extend parallel to one another through the respective passages 30, 32 in order to span a desired percentage of the width of each passage.

[0053] Turning to FIGS. 6-7, when the transformer 10 is assembled, the primary winding 40, secondary winding portions 42A, 42B, insulating assembly 100, and cooling structure 160 all extend through the passages 30, 32 of the core 20 in a precisely tailored manner. To this end, the legs 124 of the edge insulator 120 extend through opposite lateral sides of the passages 30, 32 and abut the interior of both halves 22, 24 of the core 20. In this manner, the support rails 126, 128 are aligned with and extend parallel to the respective shelves 38 on the bobbins 36. The insulating sheets 102, 112 extend through the core 20 with the openings 104 receiving the bobbins 36 and the openings 114 receiving the shelf 38 on each bobbin 36.

[0054] The primary winding 40 is positioned between both the shelves 38 on the bobbins 36 but also between the support rails 126, 128 on both legs 124 due to the openings 58, 78 in the PCBs 50, 70. This positions the primary winding 40 between a pair of the first insulating sheets 102. A pair of second insulating sheets 112 is provided on each side of that stack, and a second pair of first insulating sheets 102 is then provided on each side of that stack. The insulating assembly 100—in particular the sheets 102, 112—extends beyond the entire combined length of the respective secondary winding portions 42A, 42B and, thus the entire interface between the primary windings 40 and the secondary winding portions is electrically insulated.

[0055] The bases 52, 72 of the primary winding 40 extend between the cover members 142, 144 and within the end cap insulator 140. The tabs 55, 75 of the primary winding 40 extend laterally through the space between the cover members 142, 144 to positions outside the end cap insulator 140.

[0056] The secondary winding portions 42A, 42B, on the other hand, are provided outside the insulating assembly 100. One secondary winding portion 42A abuts the exterior of the outermost first insulating sheet 102 abutting the shelf 38 and the top (as shown) of the support rail 126. The other secondary winding portion 42B abuts the exterior of the outermost first insulating sheet 102 abutting the shelf 38 and the bottom (as shown) of the support rail 128. The openings 58, 78 allow the secondary winding portions 42A, 42B to exhibit this configuration. That said, the insulating assembly 100 and bobbins 36 cooperate to electrically insulate the primary and secondary windings 40, 42 from one another and electrically insulate the windings from the core 20. Furthermore, the insulating assembly 100 helps to increase

the creepage distance between the primary winding 40 and each respective secondary winding portions 42A, 42B.

[0057] The cooling structures 160 are provided within the passages 30, 32 and associated with each secondary winding portion 42A, 42B. In particular, one cooling structure 160 has one pass 172a extending through the passage 30 while the other pass 172b extends through the other passage 32, with both passes being substantially in the same plane and atop the same, top (as shown) secondary winding portion 42A. Similarly, the other cooling structure 160 has one pass 172a extending through the passage 30 while the other pass 172b extends through the other passage 32, with both passes being substantially in the same plane and atop the same, bottom (as shown) secondary winding portion 42B.

[0058] Due to this construction, a single cooling structure 160 provides a back-and-forth fluid path through one passage 30 and then another back-and-forth fluid path through the other passage 32. It will be appreciated that the cooling structure 160 can be configured such that multiple, consecutive passes 172 extend through the same passage 30 and/or the passage 32. Regardless, the ends 142, 144 of the cooling structures 160 are positioned at the same end of the transformer 10.

[0059] With that in mind, the straight portions 166 of each cooling structure 160 are aligned with and extend over the array of openings 54 in the secondary winding portions 42A, 42B. As shown, straight portions 166 are in close proximity to the openings 54 in the secondary winding portions 42A, 42B with no structure therebetween. The straight portions 166 of each cooling structure 160 are also aligned with the openings 54 in the primary winding 40. The secondary winding portions 42A, 42B and insulating assembly 100—specifically the second insulating sheets 112—extend directly between the straight portions 166 and the openings 54 in the primary winding 40.

[0060] The curved portions 168 extend into the end cap insulator 140 (FIGS. 8-9B). In particular, the curved portions 168 of one cooling structure 160 are positioned on one side of the projection 148 and the curved portions 168 of the other cooling structure are positioned on the other side of the projection. In other words, the cooling structures 160 extend into the respective cover members 142, 144.

[0061] Each cooling structure 160—specifically the straight portions 166—can be electrically insulated from the associated secondary winding portions 42A, 42B with a thermal enhancement material (see 161 in FIG. 12A). The thermal enhancement material 161 can be formed from, for example, thermally conductive silicone compounds or phase change materials. It will be appreciated that in lieu of [or in addition to] the thermal enhancement material 161 the PCBs 50 adjacent the cooling structure 160 can be configured such that the copper traces thereon are provided on the side of the PCB facing away from the cooling structure. In this sense, the PCB 50 itself [or rather its thickness] acts as an insulating layer between the cooling structure 160 and secondary winding portions 42A, 42B. In any case, the clips 28 help to load the cooling structure 160 against both insulating assembly 100 as well as the core 20.

[0062] The DAB module 501 includes a transistor transistor 250 provided in each of the aligned openings 54 within the primary winding 40 and each secondary winding portion 42A, 42B. As an example, the transistors transistor 250 can each be provided as a metal-oxide semiconductor field effect transistor (transistor) having a transistor die assembly that

includes one or more transistor devices. The transistors **250** can be fabricated as SiC MOSFETs, but could instead be any of a variety of different types of transistor devices, such as gallium nitride (GaN) devices, other silicon transistor devices, or silicon insulated-gate bipolar transistors (IGBTs). As an example, the transistors **250** can be either bare die components or can have a minimal surrounding package to allow a low inductance and a high thermal conductance connection to the insulating assembly **100**. Background information related to an example of the transistor **250** can be found in U.S. Pat. No. 7,119,447, the entirety of which is incorporated by reference.

[0063] Turning to FIGS. **10A-10C**, the transistor **250** shown includes an interposer **254** and a bare die **256** stacked atop one another. The bare die **256** can be formed from silicon carbide. The interposer **254** is positioned between the bare die **252** and a mounting surface, e.g., on one of the PCBs **50** and/or PCBs **70** for helping to transition the thermal coefficient of expansion and make mounting transistor **250** mounting more compatible with PCB manufacturing.

[0064] In the example of FIG. **10A**, the interposer **254** includes one or more electrical vias that extend through the interposer **254** to couple a gate terminal of the bare die **252** to a gate pad **255** and to couple a source terminal of the bare die **252** to a source pad **257** for providing electrical connectivity to the associated electrical traces of the mounting surface. The interposer **254** can be made of a material having a coefficient of expansion within about 5 ppm/K of the coefficient of expansion of the transistor **250**. Example materials for the interposer **254** include aluminum (III) oxide (Al_2O_3), aluminum nitride, and silicon nitride, and silicon dioxide.

[0065] The transistors **250** are incorporated into the primary and secondary windings **40**, **42** on the same PCBs **50**, **70** and using the single insulating assembly **100**. The transistors **250** associated with the primary side of the DAB module **501** are located next to the terminals of the primary winding **40**. The transistors **250** can have the same ground wall insulator to isolate the transistors **250** and connect to the transformer **10** on the same PCB **50**, **70** as at least one of the primary windings **40** on the transformer **10**.

[0066] A heat spreader **260** includes a base **262** and a pair of legs **264** that cooperate to define a recess **266** for receiving the interposer **254** and bare die **256**. In addition, the heat spreader **260** can correspond to a drain connection for the transistor **250**, such that the heat spreader is electrically connected to the drain terminal of the bare die **256**, e.g., on a surface of the bare die **256** opposite the interposer **254**. The heat spreader **260** is also electrically connected to the mounting surface via respective drain pads **265** arranged on the respective legs **264** and extending parallel to one another. That said, the drain pads **265** can be co-planar with one another. The heat spreader **260** can be made of a material having a coefficient of expansion within about 5 ppm/K of the coefficient of expansion of the transistor **250**. Example materials for the heat spreader **260** include molybdenum, tungsten, copper-tungsten, copper-molybdenum, and aluminum-graphite. That said, the components **252**, **254**, **260** can be connected to one another and to the PCB **50**, **70** with a sintered material made from silver or a silver alloy.

[0067] During operation, a cooling medium is supplied to the first ends **162** of the cooling structures **160**. The cooling medium can be, for example, water, a water mixture, glycol,

a glycol mixture or combinations thereof. An ethylene glycol or propylene glycol additive can be included to help prevent freezing and fouling. In any case, the cooling medium flows through each pass **172a**, **172b** in succession, passing back-and-forth through each passage **30**, **32** in the core **22**.

[0068] As noted, the flattened portions **166** are aligned with and in close proximity to the transistors **250** in the secondary winding portions **42A**, **42B**. On the other hand, heat from the primary winding **40** passes through one or more the layers of its PCBs **50**, **70** and all the PCBs **50**, **70** in the secondary winding portions **42A**, **42B**, plus the insulating assembly **100**, before reaching the coolant in the cooling structures **160**. Consequently, a material with a thermal conductivity greater than the thermal conductivity (~ 0.25 W/m-K) for flame retardant, woven glass-reinforced epoxy resin (FR4) is selected for the PCBs **50**, **70**.

[0069] The cooling structures shown and described herein provide a greater thermal conduction pathway connection to the windings and core due to their flattened profile. This profile also advantageously helps to reduce the space taken up by the cooling structures in the core passages while increasing the coolant Reynolds number, thereby increasing the fluid convection coefficient. Additionally, forming the cooling structure to pass back-and-forth through the same passage allows for both the tube and the coolant therein to be electrically conductive, which reduces construction costs and helps to improve operational reliability.

[0070] A series of spacers **180**, **181**, **182** are provided in the DAB module **501** for helping to protect the components therein and provide a more structurally sound and compact assembly. With this in mind, a pair of spacers **180** is provided for each secondary winding portion **42A**, **42B** on opposite sides of the core **20**, i.e., four total spacers **180** are provided. Each spacer **180** extends from the core **20** to the end of the straight portions **166** of the cooling structure **160** of that particular one of the secondary winding portions **42A**, **42B**. That said, the straight portions **166** are sandwiched between [but still spaced from] the outermost PCB **50** of that particular one of the secondary winding portions **42A**, **42B** and the spacer **180** associated therewith.

[0071] A spacer **181** (FIGS. **7** and **9B**) is provided between the curved portions **168** of each secondary winding portion **42A**, **42B** and the end wall **146** of the end cap insulator **140**. The spacers **181** extend to opposite sides of the projection **148** on the end wall **146**. A spacer **182** (FIG. **10B**) is also provided in the plane of the primary winding **40** and on the opposite side of the edge insulator **120** from the primary winding.

[0072] PCBs **190** are provided on both sides of the core **20** and extend along the length **L** of the DAB module **501** parallel to one another. The PCBs **190** receive the support members **80** extending from the tabs **75** on the primary winding **40** to secure the two together and laterally space the PCBs **190** from the secondary winding portions **42A**, **42B**. That said, capacitors **192** are connected to each PCB **190** on opposite lateral sides of the secondary winding portions **42A**, **42B** and opposite longitudinal sides of the core **20**.

[0073] The PCBs **190** each include openings for receiving a second core **200** that acts as a transformer of an isolated DCDC converter to provide control and gate-drive power from the secondary side to the primary side (or vice versa). The second cores **200** are made of a ferritic material and extends to opposite sides of the PCB **190**. Additional capaci-

tors **192** are provided in the lateral space between the spacer **182** and each PCB **190**. Additional capacitors **194** can be provided on the tabs **55** of the primary winding **40** (FIG. 9).

[0074] The transistors **250** associated with the primary side of the DAB module **501** are located next to the terminals of the primary winding **40**. The DAB module **501** can include operating power electronics, such as including gate drive power supplies, gate driver integrated circuits (ICs), voltage and current sense circuits, and switch control signals from a controller. Because there are active components on each side of the transformer isolation barrier formed by the transformer **10**, the DAB module **501** can include features for transferring power and logic signals across the isolation barrier. FIG. 8 shows one possible configuration that includes a long planar transformer core **20** to transfer power, and fiber optic links **210** between logic circuits to transfer logic signals across the isolation barrier.

[0075] Once the transformer **10** is assembled, the entire assembly is potted in a thermally conductive, electrically insulating potting compound **300** (FIG. 11) that leaves the ends **162**, **164** of the cooling structures **160** exposed/accessible. Example materials for the potting compound **300** include nylon, silicone, PPS, PPA, PBT, epoxy, or polyester. In any case, the potting compound **300** should have a low durometer (<60 Shore 00) and a low un-cured viscosity (<5000 cPs), inserted under a vacuum.

[0076] Referring to FIG. 12A, it will be appreciated that the transformer **10** can be modified to be a stand-alone assembly, i.e., one not implemented into the DAB module **501**. In such a configuration, the PCBs **50**, **70** are modified to omit the openings **54** for the transistors **250**. Additional DAB module **501** components, such as the PCBs **190**, capacitors **192**, core **200**, and fiber optic links **210** are also omitted. That said, the stand-alone transformer **10** can also be potted as shown in FIG. 12B.

[0077] To this end, a case **310** cooperates with the potting **300** to increase the durability of the transformer **10**. The case **310** should be made of plastic, have a high temperature rating, and a high CTI greater than about 600V. The case **310** includes one or more compression clips **312** positioned along the length *L* of the potting **300** for providing a sustained, compressive force on the potting while also helping to mitigate any partial discharge in the potting that may occur during the life of the transformer **10**. The use of silicone for the potting **300**, in combination with the clips **312**, can help provide a self-healing function for the transformer **10**.

[0078] It will be appreciated that the case **310** can include one or more electronic couplings or connections **313** that enable the transfer of power from/between exterior of the potting **300** to the interior thereof. The connections **313** can be provided on both ends of the transformer **10**/DAB module **501**. That said, the connections **313** can also be provided on both ends of the case **310** shown in FIG. 11.

[0079] The example of FIG. 13 illustrates an example of a circuit diagram of a DAB circuit **400** associated with the transformer **10** and DAB module **501** described herein. That said, the DAB circuit **400** is demonstrated as a high-frequency DC-to-DC converter that includes a transformer **402** with a series inductor, a first stage **404** that includes a full active bridge, and a second stage **406** that likewise includes a full active bridge. In the example of FIG. 13, the transformer **402** is demonstrated as including an inductor in series with the primary winding. The inductor can be a

separate inductor, or can correspond to a leakage inductance of the transformer **402** if the leakage inductance is sufficient to not require a separate inductor.

[0080] The first stage **404** is demonstrated as an input stage in which the switches of the full active bridge are switched to provide a current through the primary winding of the transformer **402** from an input voltage V_{IN} across an input capacitor C_1 . Similarly, second stage **406** is demonstrated as an output stage in which the switches of the full active bridge are switched to provide an induced current through the secondary winding of the transformer **402** to provide an output voltage V_{OUT} across an output capacitor C_2 . The DAB circuit **400** can be provided in a high-power voltage converter to provide for bidirectional power flow, low modulation complexity, case of resonant conversion, e.g., because it can be operated at fixed 50% duty cycle, and soft-switching, and high conversion efficiency.

[0081] The DAB circuit **400** is demonstrated in the example of FIG. 13 and described above as unidirectional. However, it is to be noted that the description of the DAB circuit **400** as converting a DC voltage at the input stage via the primary winding to provide a DC voltage at the output stage via a secondary winding is one example implementation regarding the orientation of the DAB circuit **400**. As another example, the DAB circuit **400** can operate bidirectionally, such that the input stage/primary and output stage/secondary could be reversed. The potential bidirectional operation of the DAB circuit **400** can likewise be applicable to the further examples described below.

[0082] As an example, the transformer **10** can be implemented as the transformer **402** in the DAB circuit **400** to achieve high-power voltage conversion in a compact form-factor, such as the DAB module **501**, as described herein. The example of FIG. 14 illustrates another example of a circuit diagram of a DAB circuit **500**. The DAB circuit **500** is demonstrated similar to the DAB circuit **400** in the example of FIG. 13, and thus includes a transformer **502**, a first stage **504** that includes a full active bridge, and a second stage **506** that likewise includes a full active bridge. However, the first stage **504** also includes an additional full active bridge, such that the DAB circuit **500** can provide AC-DC conversion. Similar to as described above, the transformer **10** can be implemented as the transformer **502** in the DAB circuit **500** to achieve high-power voltage conversion in a compact form-factor. Furthermore, the DAB circuit **500** can correspond to the DAB module **501** described above in the example of FIG. 1.

[0083] For example, the switches of the first and second stages **504** and **506** can correspond to the transistors **250** described above, and the capacitors C_1 and C_2 can correspond to the capacitors **190** and **192** described above. In addition, the DAB circuit **500** includes a set of controls for operating the DAB circuit **500**. The controls include a master controller **508**, a set of isolated communication logic **510**, low-side logic **512**, and high-side logic **514**. As an example, the master controller **508** and/or the communication logic **510** can be external to the DAB circuit **500**, and thus external to the DAB module **501**, e.g., to control all DAB modules **501** in a given set of DAB modules, as described in greater detail below. The isolated communication logic **510**, secondary-side logic **512**, and primary-side logic **514** can be included in the DAB module **501**. For example, the fiber-optic link **210** of the DAB module **501**

can be implemented to provide logic signals between the primary-side logic 514 and the secondary-side logic 512.

[0084] To achieve conversion of the medium amplitude AC voltage to the high amplitude DC voltage, a group of DAB circuits 500 can be electrically combined to form a voltage converter circuit. The example of FIG. 15 demonstrates a voltage converter circuit 700 that includes a plurality of DAB circuits 702, demonstrated as twelve in the example of FIG. 15, that provide voltage conversion for an EV charging station 704. As an example, each of the DAB circuits 702 can correspond to the DAB circuit 500 described above to collectively convert a medium amplitude AC voltage, such as at least 5 kVAC (e.g., 7.5 kVAC), provided from an AC input 706 to a high amplitude DC voltage, such as at least 750 VDC (e.g., 1000 VDC), provided to the EV charging station 704. Each of the DAB circuits 702 includes an input (primary) stage 708 and an output (secondary) stage 710 that are interconnected by a high-frequency transformer 712, e.g., the planar transformer 10 described above.

[0085] In the example of FIG. 15, the primary windings of the transformers 712 of each of the DAB circuits 702 are connected in series via the input stage 708, and the secondary windings of the transformers 712 of each of the DAB circuits 702 are connected in parallel via the output stage 710. As a result, the voltage converter circuit 700 exhibits a series-in parallel-out architecture. By placing the primary windings of the transformer 712 in series and the secondary windings of the transformer 712 in parallel, a large voltage step-down can be achieved using approximately 1:1 turns ratio of the transformers 712. It is noted that turns ratios other than 1:1 can instead be implemented, e.g., between approximately 0.5:1 and 2:1. This general topology allows for isolated voltage conversion without the use of large, e.g., 50 or 60 Hz, transformers. A higher transformer frequency allows for the use of smaller transformers, such as the transformer 10. Similar to as described above, while the term “primary” refers to the AC voltage side and the term “secondary” refers to the DC side, the configuration of the voltage converter circuit 700 can operate bidirectionally, such that the primary and secondary could be reversed in practice.

[0086] In the example of FIG. 15, the voltage converter circuit 700 also includes an inductor L_{IN} in series with the AC input 706, thereby rendering the front end of the voltage converter circuit 700 to act as an active boost rectifier providing the DC voltage on the capacitor C_1 in the input stage 708 of each of the DAB circuits 702. The combination of the AC-DC DAB circuits 702 having the inputs connected through the primary winding of each of the transformers 712 in series provides for a multilevel converter input stage which allows multiple lower voltage transistors to be used as the switches of the input stages 708 to accommodate a high voltage input from the AC input source 706.

[0087] For example, as described above, the transistors of the input and/or output stages 708 and 712 can be arranged as the transistors 250 of the DAB module 501. The series inductor L_{IN} for the series connected modules can make the input stages 708 of the voltage converter circuit 700 operate as a cascaded active boost rectifier. While the single inductor L_{IN} is demonstrated in the example of FIG. 15 in series with the series-connected input stages 708, an individual inductor can instead be included in the input stage 708 of each of the DAB circuits 702, and thus included in the DAB module

501, such as to provide for a more modular construction of the voltage converter circuit 700.

[0088] The voltage converter circuit 700 can each be configured in a variety of different ways to provide AC-DC conversion, DC-DC conversion, DC-AC conversion, or AC-AC conversion. For example, an additional full active bridge can be provided to the output of each of the DAB circuits 700 to provide an AC output voltage, e.g., 600 VAC, collectively. As another example, the secondary windings of the transformers 712 in the output stages 710 of the DAB circuits 700 may be coupled together in different ways in a given one of the voltage converter circuits 802. For example, for an even number of DAB modules 700, pairs of the output stages 710 of respective DAB circuits 702 can be arranged in series instead of parallel to provide twice the output voltage, e.g., 2 kV, at half the output current than the fully parallel connection shown in the example of FIG. 15.

[0089] Similarly, pairs of the input stages 708 of respective DAB circuits 702 can be arranged in parallel instead of series to provide twice the input current over a uniform AC voltage for each pair than the fully series connection shown in the example of FIG. 15. The voltage converter circuit 700 can thus be configured with any variation of AC or DC input or output, and any combination of series and parallel couplings of the input stages 708 and the output stages 710 to affect the amplitude of the output voltage V_{OUT} .

[0090] The example of FIG. 16 demonstrates another example of an EV charging system 800. The EV charging system 800 includes three separate phases of AC power, Phases A through C, that are provided to three respective voltage converter circuits 802. Each of the voltage converter circuits 802 can correspond to the voltage converter circuit 700, and can thus each include a plurality of DAB circuits 500, e.g., that each correspond to the DAB module 501. Accordingly, each of the voltage converter circuits 802 can implement the same DAB circuits 500 for each of the voltage converter circuits 802, thereby providing for a simple modular design of each of the voltage converter circuits 802.

[0091] In the example of FIG. 16, the three separate phases of AC power are provided from AC power sources 804 to provide 13 kVAC three-phase power to the three voltage converter circuits 802, such that each of the voltage converter circuits 802 can generate 1000 VDC. As an example, the 13 kVAC three-phase power can be connected in a wye configuration to provide the 7.5 kVAC to each of the voltage converter circuits 802, thus reducing the number of DAB circuits 500 in each of the respective voltage converter circuits 802. Alternatively, the 13 kVAC three-phase power can be connected in a delta configuration to provide the 13 kVAC to each of the voltage converter circuits 802.

[0092] The EV charging system 800 includes a master controller 806 that can draw sinusoidal current in phase with the source phase voltages to achieve near-unity power factor, i.e., power correction (PFC). As an example, the master controller 806 can regulate the output voltage on the capacitors C_2 of each of the DAB circuits 500, while simultaneously maintaining an approximately equal voltage across all the capacitors C_1 in each of the DAB circuits 500. Therefore, the master controller 806 can provide active voltage balancing in providing the output voltage to each of a plurality of isolated EV chargers for the EV charging system 800.

[0093] For example, the master controller 806 can control the switches of the first and second stages 504 and 506 between the capacitors C_1 and C_2 of each of the DAB circuits 500 to perform the sinusoidal current control in conjunction with regulating the voltage on the respective capacitor C_2 . In the example of FIG. 16, the three phases of the AC power 804 are balanced, and all of the capacitors C_2 of each of the DAB circuits 500 are connected in parallel across the three respective power converter circuits 802. Therefore, the three balanced phases of the AC power 804 can result in minimization of the total amount of capacitance of the capacitors C_2 . The cascaded AC/DC front end stages of each of the DAB circuits 500 in each of the voltage converter circuits 802 act to maintain approximately balanced and equal voltages on the capacitors C_1 by directing the input AC current in or out of each of the capacitors C_1 as needed for proper regulation.

[0094] What have been described above are examples of the present invention. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the present invention, but one of ordinary skill in the art will recognize that many further combinations and permutations of the present invention are possible. Accordingly, the present invention is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims.

What is claimed is:

1. A voltage converter circuit comprising a plurality of dual active bridges (DABs), each of the DABs comprising:
 - a first stage comprising a plurality of first switches;
 - a second stage comprising a plurality of second switches;
 - and
 - a transformer comprising a first winding coupled to a least one of the first switches and a second winding coupled to at least one of the second switches;
 wherein at least one of the first and second stages of a plurality of the DABs that is a proper subset of the DABs are arranged in parallel with respect to the first winding of the transformer in each of the respective plurality of the DABs, with the parallel first stages of the plurality of the DABs being arranged in series with

respect to the first winding of the transformer of at least one other DAB of the DABs, or

wherein at least one of the first and second stages of a plurality of the DABs that is a proper subset of the DABs are arranged in series with respect to the second winding of the transformer in each of the respective plurality of the DABs, with the series second stages of the plurality of the DABs being arranged in parallel with respect to the second winding of the transformer of at least one other DAB of the DABs.

2. The circuit of claim 1, wherein the first stage is configured to receive a direct current (DC) input voltage, wherein the first stage comprises an input bridge portion comprising the first switches configured to provide the DC input current to the primary winding of the transformer.

3. The circuit of claim 1, wherein the first stage is configured to receive an alternating current (AC) input voltage, wherein the first stage comprises:

- a first input bridge portion comprising a first set of the first switches configured to convert the AC input voltage to a direct current (DC) input voltage; and

- a second input bridge portion comprising a second set of the first switches configured to provide the DC input current to the primary winding of the transformer.

4. The circuit of claim 1, wherein the second stage is configured to provide a direct current (DC) output voltage, wherein the second stage comprises an output bridge portion comprising the second switches configured to rectify an output current through the second winding of the transformer to provide the DC output voltage.

5. The circuit of claim 1, wherein the second stage is configured to provide an alternating current (AC) output voltage, wherein the second stage comprises:

- a first output bridge portion comprising a first set of the second switches configured to rectify an output current through the second winding of the transformer to provide a DC voltage; and

- a second input bridge portion comprising a second set of the second switches configured to convert the DC current to the AC output current.

* * * * *